

TCP/IP 프로토콜 및 인터넷

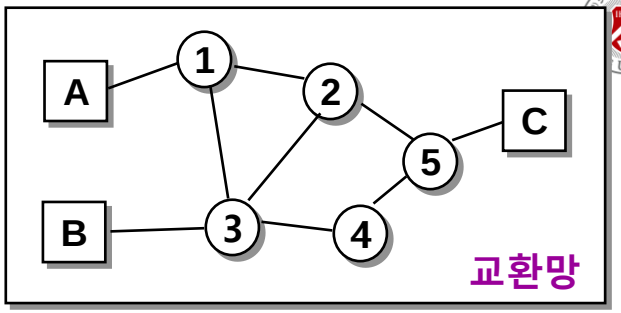
1. A Network Primer
2. TCP/IP and Related Protocols
3. TCP/IP 관련 시스템 화일 및 관리 명령어들

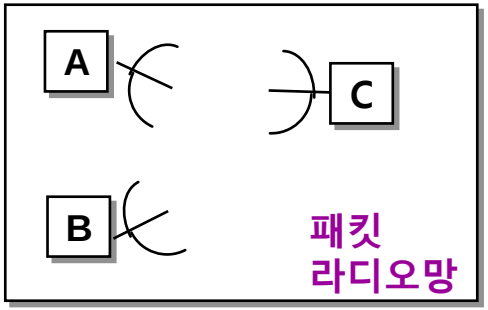
서강대학교
Sogang University

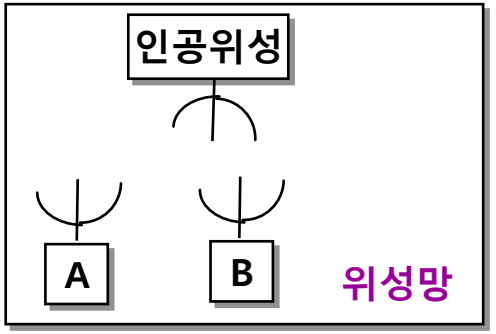
1. A Network Primer

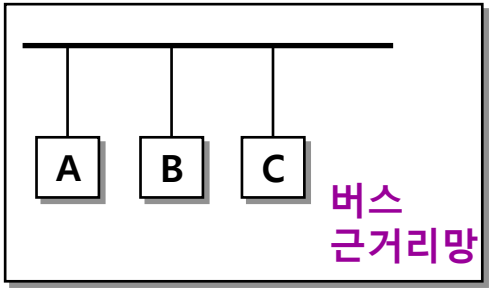
- define some basic networking terms and concepts

- 통신망의 종류
  - 교환망 (Switched Network)
    - ⇒ 데이타가 전달되기 위하여 교환기 경유
    - ⇔ 회선 교환망 (Circuit-switched Network)
      - ⇒ 전송하기 전에 경로가 먼저 설정됨 (예:전화망)
    - ⇔ 패킷 교환망 (Packet-switched Network)
      - ⇒ 각 패킷 단위로 동적 경로 설정
  - 동시 통신망 (Broadcasted Network)
    - ⇒ 모든 스테이션은 하나의 공유 와이어나 케이블로 연결됨 (예: 방송망)
    - ⇔ 패킷 라디오망 (Packet Radio Network)
    - ⇔ 위성망 (Satellite Network)
    - ⇔ 근거리망 (Local Network)
  - 컴퓨터 통신망 :
    - ⇔ 패킷 교환망 + 근거리망









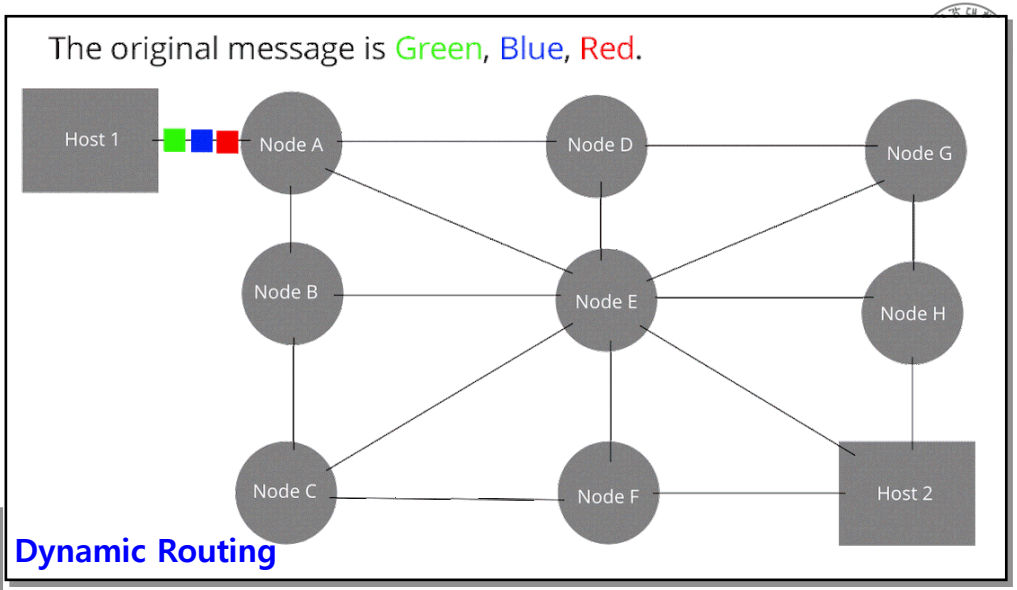
Circuit -switched Networks

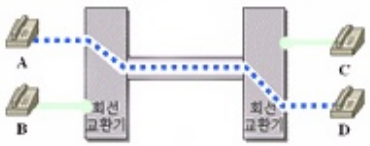
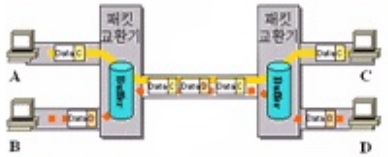
- Entire message is passed.
- There is a dedicated communication link.

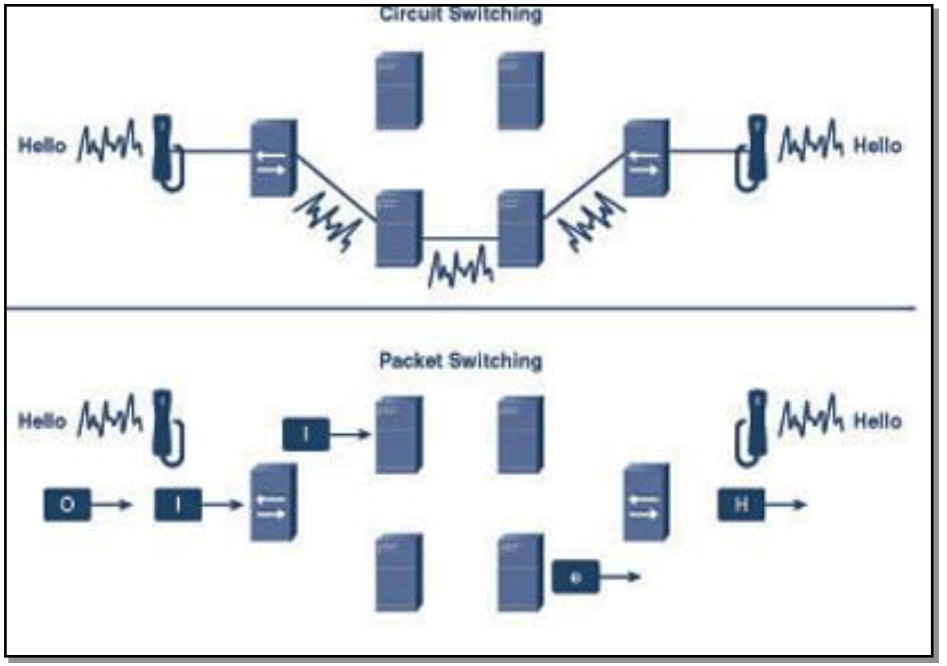
Packet-switched Networks

- The message is broken down into small packets.
- Every packet follows a different route.

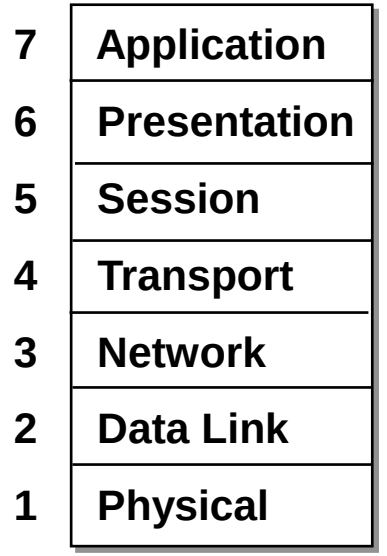
Buzzle.com

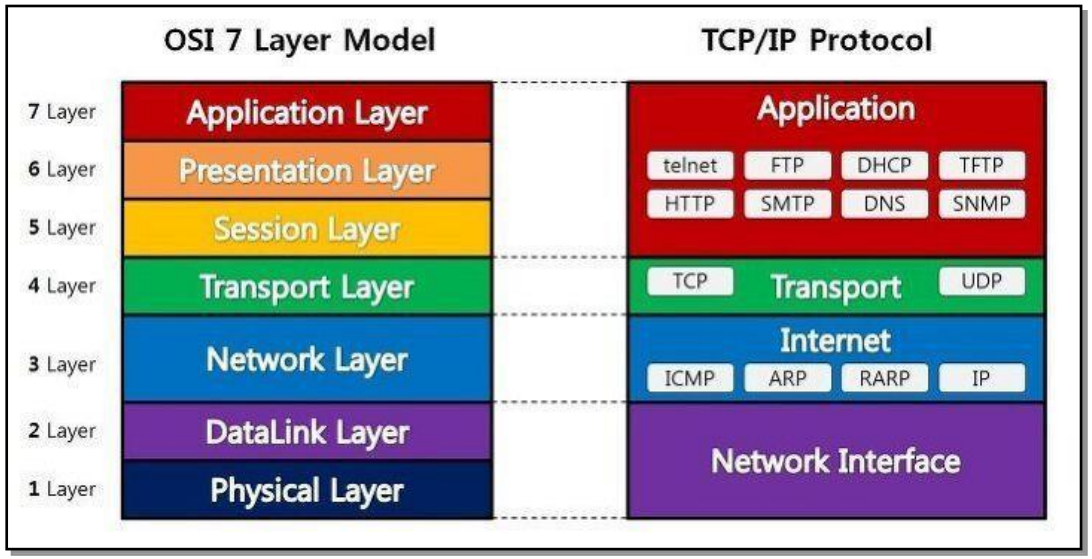


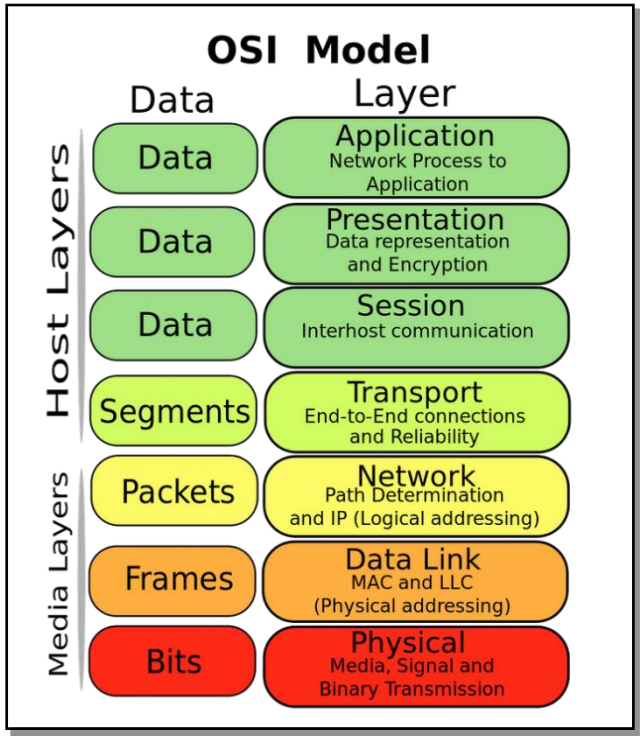
구 분	회선 교환	패킷 교환
개념도		
링크의 점유	접속이 이루어지기 전 Signaling을 통해 미리 정하고 그 회선을 통화가 종료 될 때까지 독점	사전 Signaling이 없이 여러 사용자 소유의 패킷드링 모든 가용한 링크를 따라 전달
묵음시	두 통화자 사이 대화가 없어도 회선 점유	다른 사용자가 사용함
QoS	음성과 같은 Real-time 용의 QoS 제공	Best-Effort형의 낮은 QoS제공
요금제	상세과금으로 다양한 요금제 가능	정액제, 종량제
통화량 폭주시	가용한 링크가 없을 시에는 한계를 넘는 접속을 허용하지 않음	패킷의 전달은 가능하나 병목현상으로 인한 지연과 손실이 발생할 수 있음
Routing	Static Routing (미리 정해진 경로에 의함)	Dynamic Routing(패킷마다 다른경로 선택)
망관리	구성, 장애, 성능 관리의 개념 확립	망관리의 개념이 없음

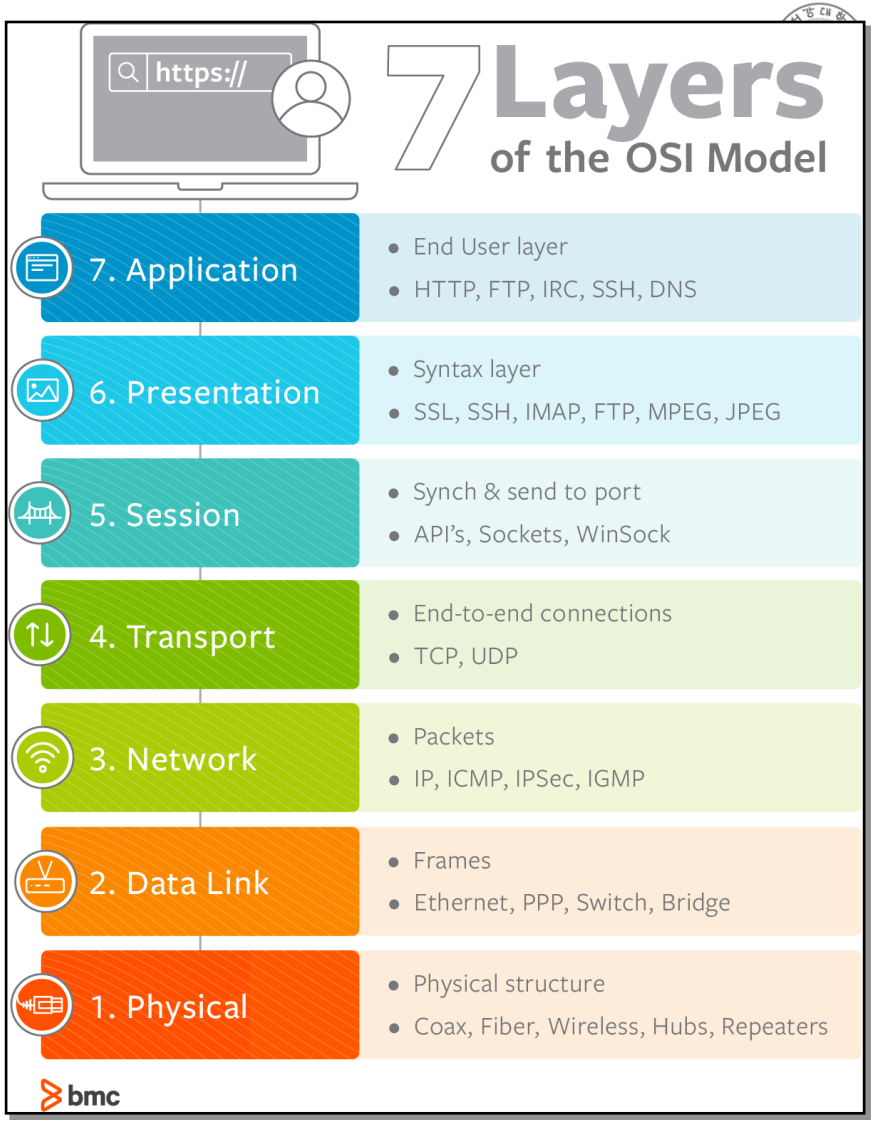


- **컴퓨터 통신망**
  - ⇔ 둘 이상의 컴퓨터가 통신망으로 연결된 경우, 이들 컴퓨터의 집합을 컴퓨터 통신망이라 한다.
  - – **Layer and Protocol**
    - ⇔ **Layer :**
    - ⇔ **Protocol :**
      - ⇒ **a set of rules and conventions between the communicating participants**
  - – **OSI 7 layer**
    - ⇔ **Application Layer**
      - ⇒ **Provides services from all layers to a user**
    - ⇔ **Presentation Layer**
      - ⇒ **Provides data transformation services**
    - ⇔ **Session Layer**
      - ⇒ **Establish and terminate connections**
    - ⇔ **Transport Layer**
      - ⇒ **End-to-end flow control**
    - ⇔ **Network Layer**
      - ⇒ **packet routing**
    - ⇔ **Data Link Layer**
      - ⇒ **error-free data delivery, point-to-point comm.**
    - ⇔ **Physical Layer**

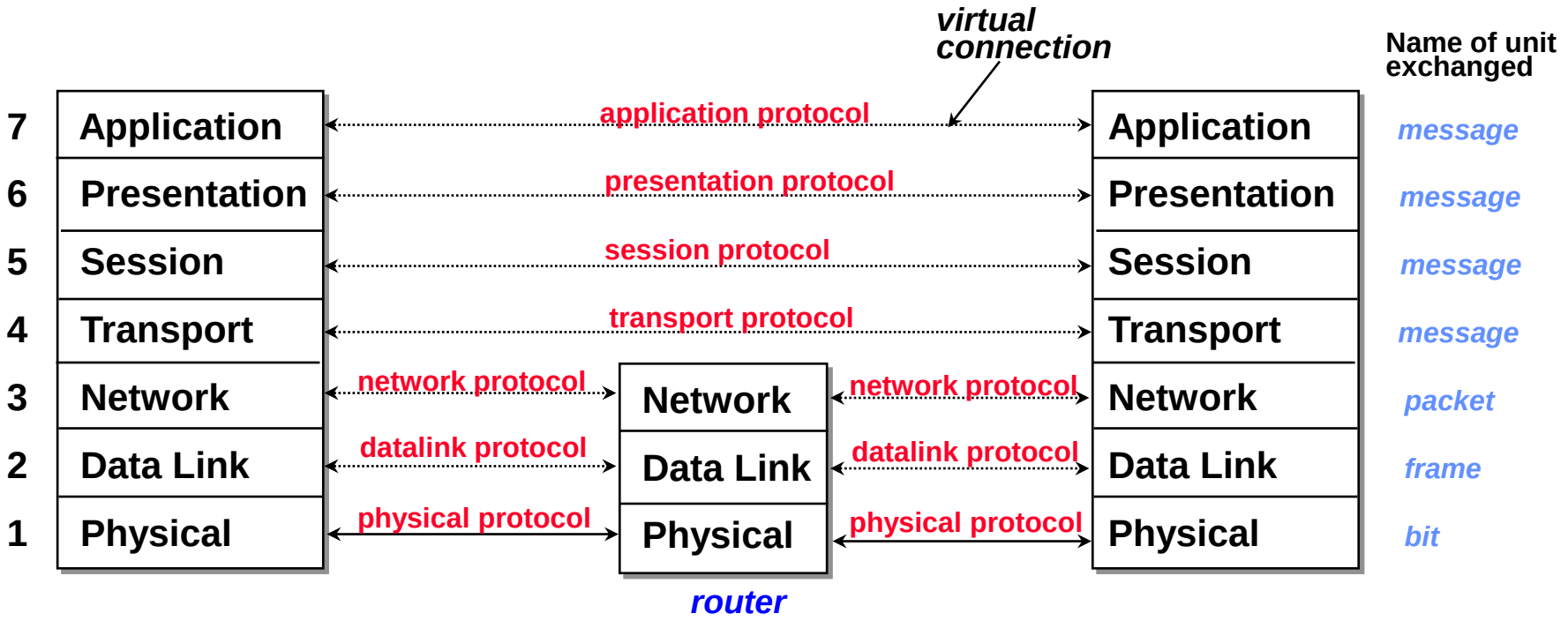



OSI 7 Layer Model
TCP/IP Protocol
7 Layer
Application Layer
6 Layer
Presentation Layer
5 Layer
Session Layer
4 Layer
Transport Layer
3 Layer
Network Layer
2 Layer
DataLink Layer
1 Layer
Physical Layer
Application
telnet
FTP
DHCP
TFTP
HTTP
SMTP
DNS
SNMP
TCP
Transport
UDP
Internet
ICMP
ARP
RARP
IP
Network Interface


OSI Model
Data
Layer
Host Layers
Media Layers
Data
Application
Network Process to Application
Data
Presentation
Data representation and Encryption
Data
Session
Interhost communication
Segments
Transport
End-to-End connections and Reliability
Packets
Network
Path Determination and IP (Logical addressing)
Frames
Data Link
MAC and LLC (Physical addressing)
Bits
Physical
Media, Signal and Binary Transmission


https://
7 Layers
of the OSI Model
7. Application
End User layer
HTTP, FTP, IRC, SSH, DNS
6. Presentation
Syntax layer
SSL, SSH, IMAP, FTP, MPEG, JPEG
5. Session
Synch & send to port
API's, Sockets, WinSock
4. Transport
End-to-end connections
TCP, UDP
3. Network
Packets
IP, ICMP, IPSec, IGMP
2. Data Link
Frames
Ethernet, PPP, Switch, Bridge
1. Physical
Physical structure
Coax, Fiber, Wireless, Hubs, Repeaters
bmc

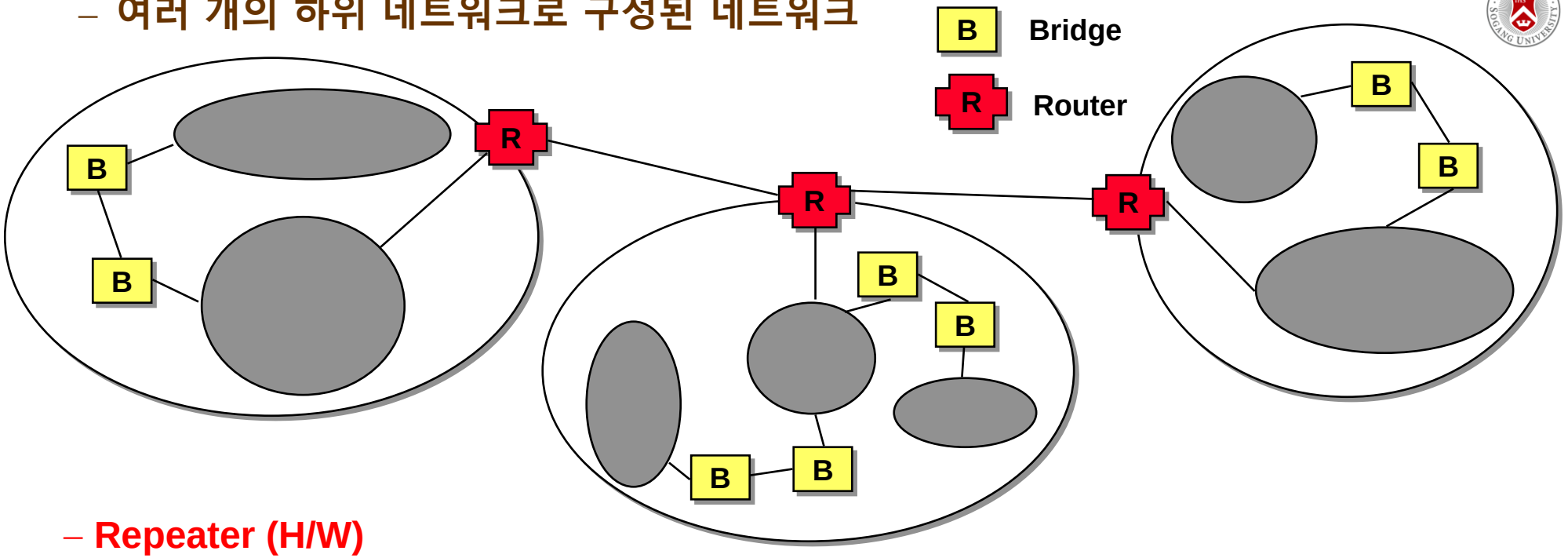
• Protocol Architecture



- Peer-to-peer protocol
- Services vs. Protocol

- Why layering ?

– 여러 개의 하위 네트워크로 구성된 네트워크



– **Repeater (H/W)**

⇔ typically just copy electrical signal from one segment of a network to the next (operate at the physical layer)

– **Bridge (HW/ or S/W)**

⇔ copy frames from one network to the next (operate at the datalink layer)

– **Router (HW/SW)**

⇔ not only moves the packet from one network to another, but it can also make decisions about what route the packet should take (operate at the network layer)

– **Gateway**

⇔ a generic term that refers to an entity used to interconnect two or more networks

– **Protocol Suit (protocol family)**

⇔ a collection of protocols from more than one layer that forms the basis of a useful network

⇔ 예) TCP/IP, Xerox NS, IBM SNA, ..

– **Network Byte Order**

⇒ not all computers store the byte that comprise a multibyte value in the same order

⇔ little endian

⇒ the low-order byte at the starting address

⇒ Intel 80x86 (IBM PC), DEC VAX, DEC PDP-11,

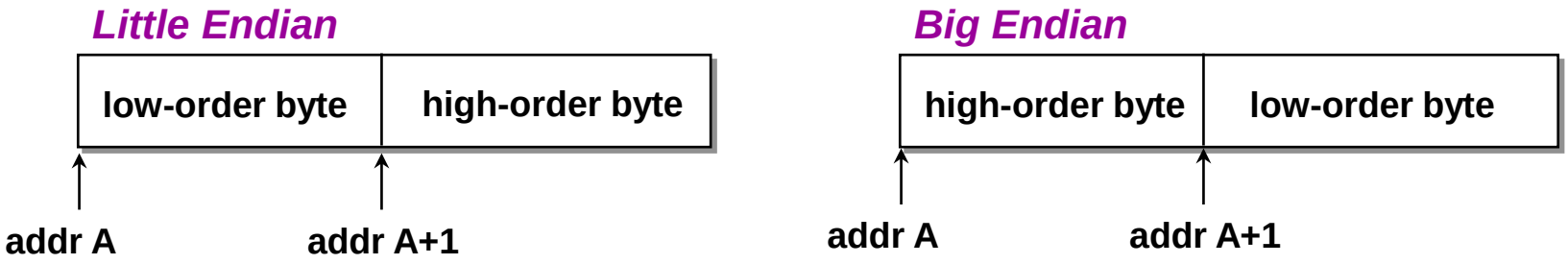
⇒ ARM (https://www.quora.com/Is-ARM-big-endian-or-little-endian)

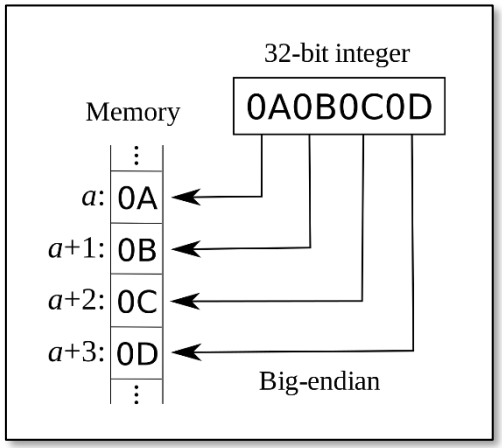
⇔ big endian

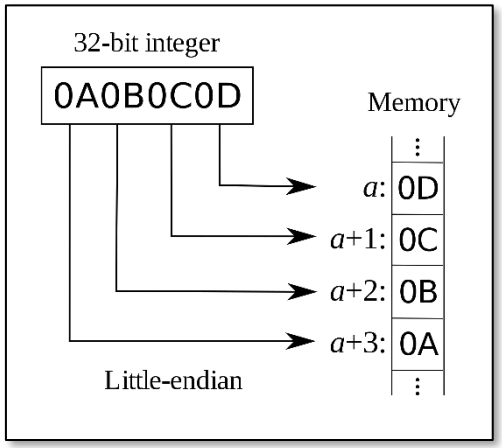
⇒ the high-order byte at the starting address

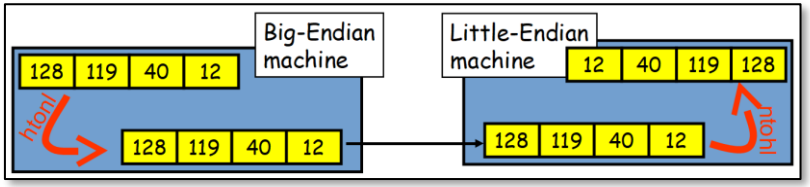
⇒ IBM 370, Motorola 68000, ...

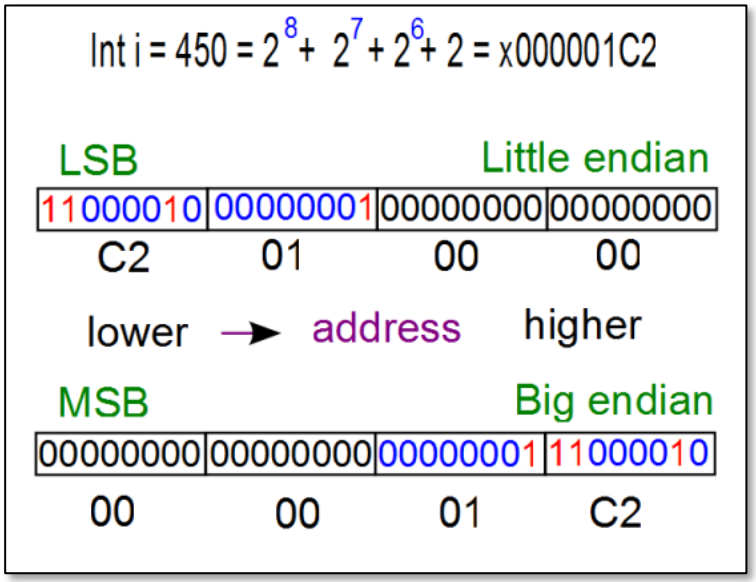
⇒ TCP/IP, XNS, SNA all use the big-endian format for 16-bit integer











Hexadecimal Number 0x 1 2 3 4

Least Significant Digit

Most Significant Digit

Broken into bytes: Most Significant 0x12

Least Significant 0x34

Big-Endian Storage

Address	Data	
0	0x12	MSByte
1	0x34	LSByte

Little-Endian Storage

0	0x34	LSByte
1	0x12	MSByte

Big-Endian Transmission

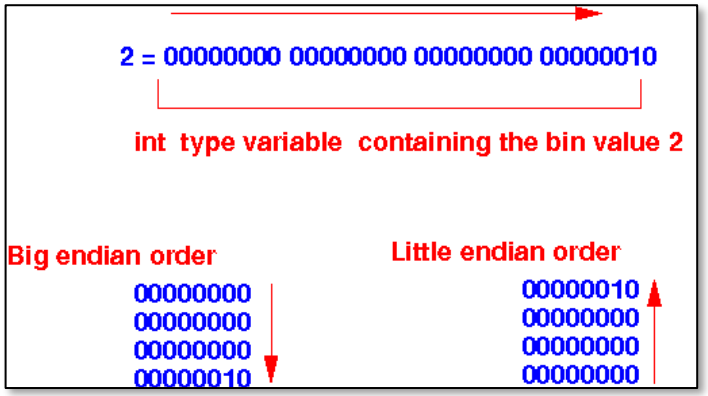
Byte Transmitted	0x12	0x34
Time	0	1

Little-Endian Transmission

Byte Transmitted	0x34	0x12
Time	0	1

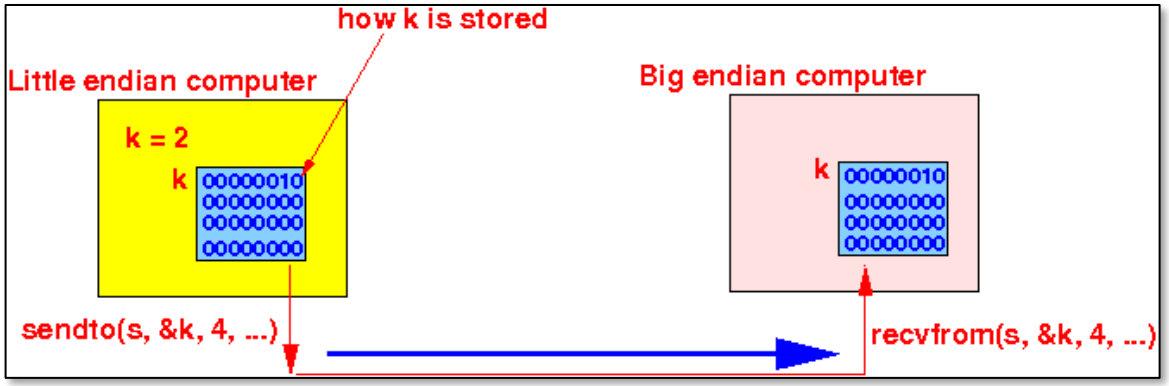
• Sending and receiving binary numbers

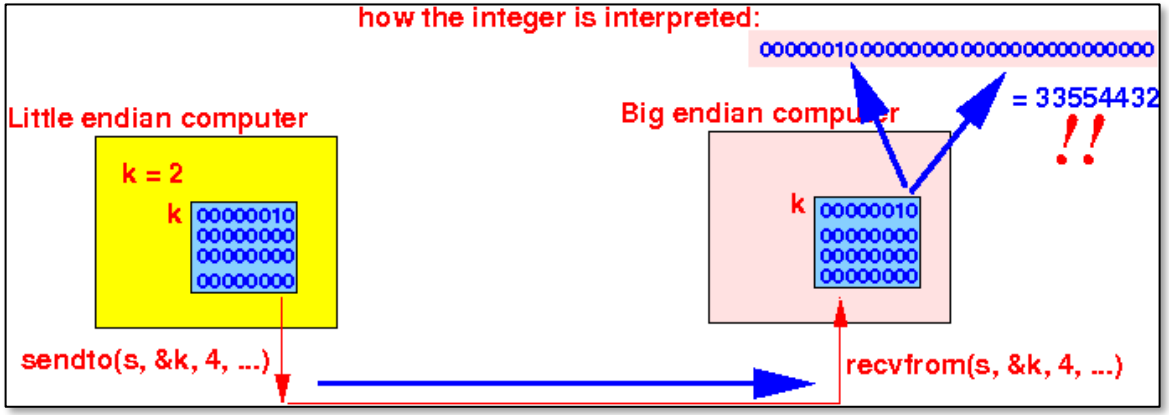
Number is stored in Multiple Bytes !!
- integer/float : 4 bytes
- char : 1 byte (ASCII)



- **String/text data** (**stored** in the ***same*** **order** in **both big endian computer** and **little endian computer**)
- **integer/float binary data** (**stored** in ***different*** **order** in **big endian computer** and **little endian computer**)

`htonl( ) and ntohl( )`





Sender

```
int k = 1;    // k is stored in host byte order

printf("Sending: int value = %d₩n", k);

int  x = htonl(k);      // Convert to network byte ordering

sendto(s,  &x, sizeof(int), 0 /* flags */,
        (struct sockaddr *)&dest_addr, sizeof(dest_addr));
```

Receiver

```
int x;

src_addr_len = sizeof(src_addr);
len = recvfrom(s, &x, sizeof(int), 0 /* flags */,
          (struct sockaddr *) &src_addr, &src_addr_len);

k = ntohl( x );      // Convert int from network byte ordering
                     // to host byte ordering

printf("k = %d", k);
```

- **Fragmentation and Reassembling**

  ⇔ most network layers have a maximum packet size that they can handle, based on the characteristics of the data-link layer that they are sending its packets (MTU)

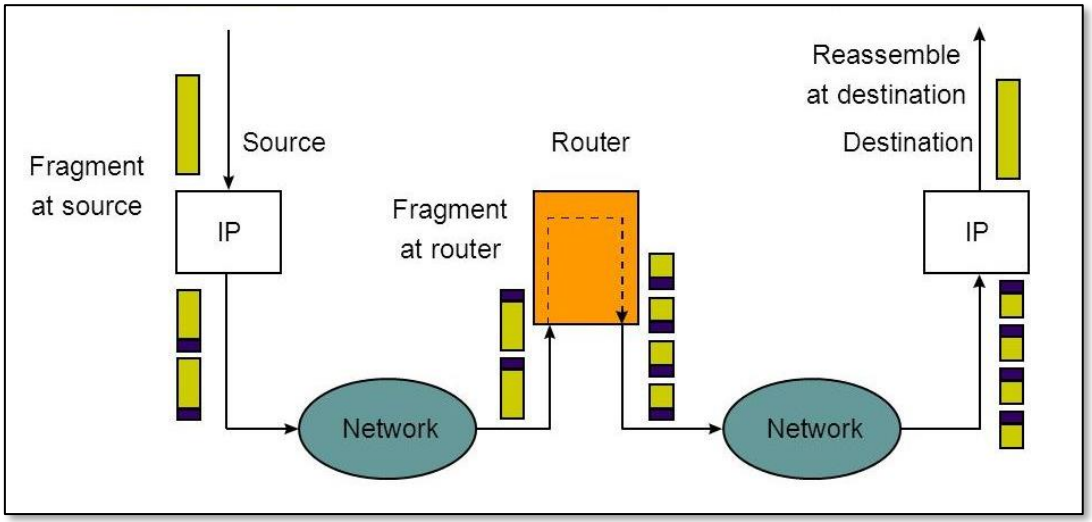
  ⇒ Ethernet : 1500 bytes, Token Ring : 4464 bytes

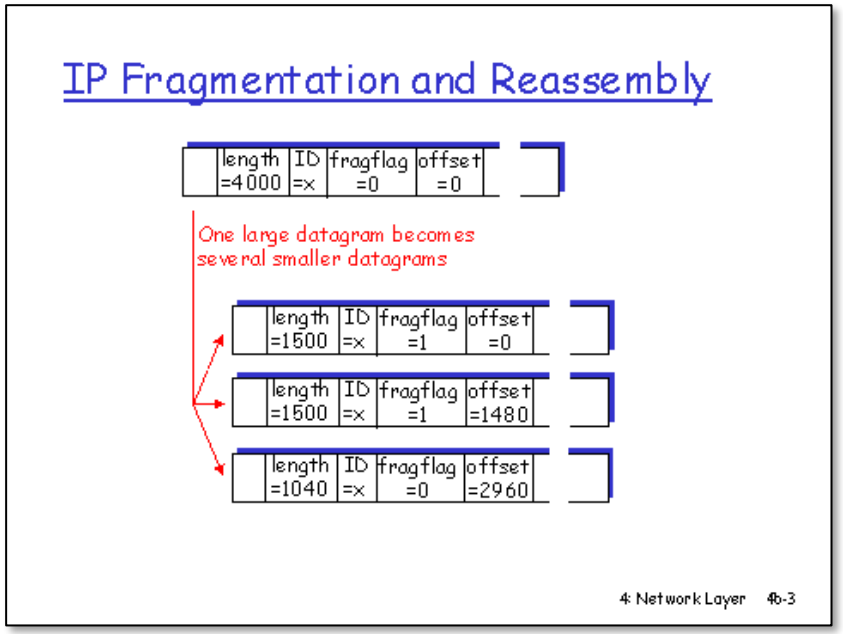
  – **fragmentation (segmentation)**

  ⇔ the breaking up a data stream into smaller pieces

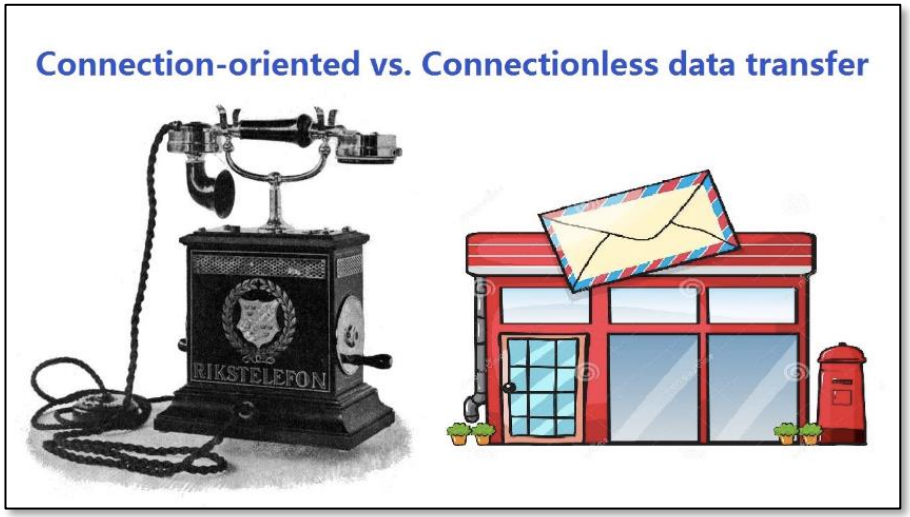
  – **reassemblying**

  ⇔ the reverse of fragmentation





- **Modes of Services (at transport layer)**
  - **Connection-oriented (virtual circuit) vs. Connectionless (datagram)**
    - ⇔ ***Connection-oriented services*** **:**
      - ⇒ **the two application program establish a logical connection with each other before communication can take place**
      - ⇒ **by TCP**
    - ⇔ ***Connectionless services*** **:**
      - ⇒ **message called datagrams are transmitted independently, and each must contain all information required for its delivery**
      - ⇒ **by UDP**



  - **Sequencing**
    - ⇔ **the property that the data is received by the receiver in the same order as it is transmitted by the sender**

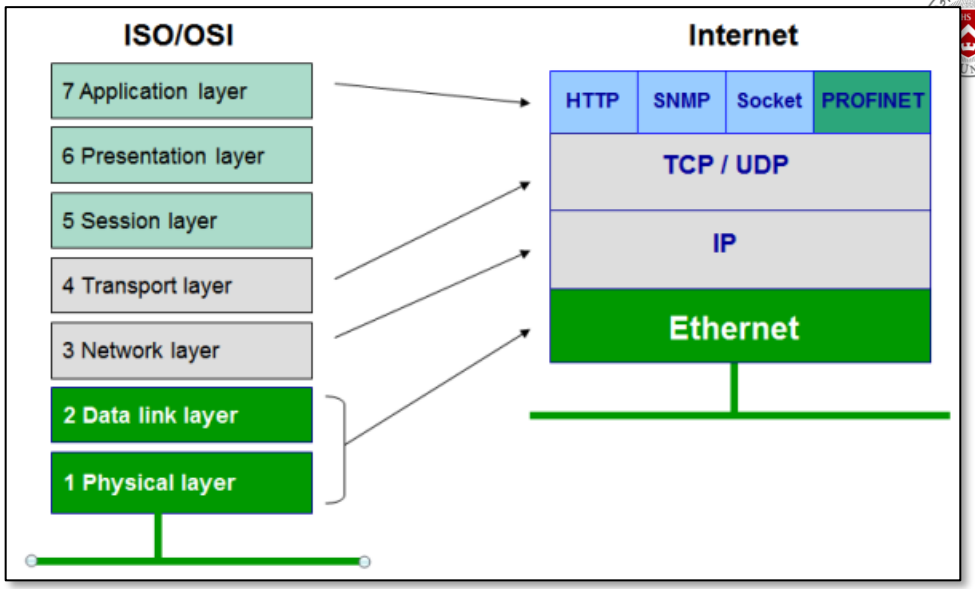
- Error Control
  - ⇔ error free data is received by the application programs
    - ⇒ the data gets corrupted
    - ⇒ the data got lost
- Flow Control
  - ⇔ the sender does not overwhelm the receiver by sending data at a rate than the receiver can process the data
  - ⇔ TCP vs. UDP
    - ⇒ TCP : provides sequencing, error control, and flow control
    - ⇒ UDP : does not provide any of these services
  - ⇔ End-to-End vs. Hop-by-Hop flow control
    - ⇒ end-to-end : regulate the flow of data between the two user processes that are exchanging data (by TCP)
    - ⇒ hop-by-hop : worries about the flow of data at each hop of the comm. link (By Network)
- Buffering and Out-of-Band Data

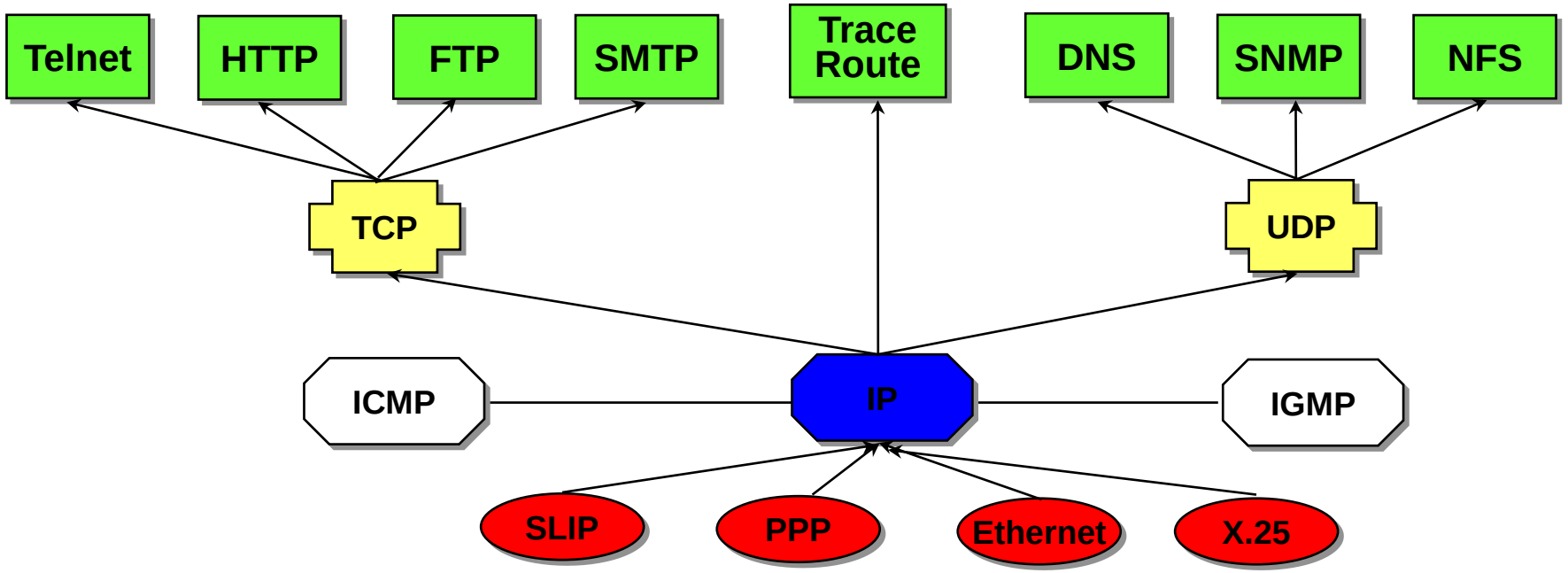
• Association
- specifies the two processes that make up a connection
- {protocol, local-addr, local-process, foreign-addr, foreign-process)

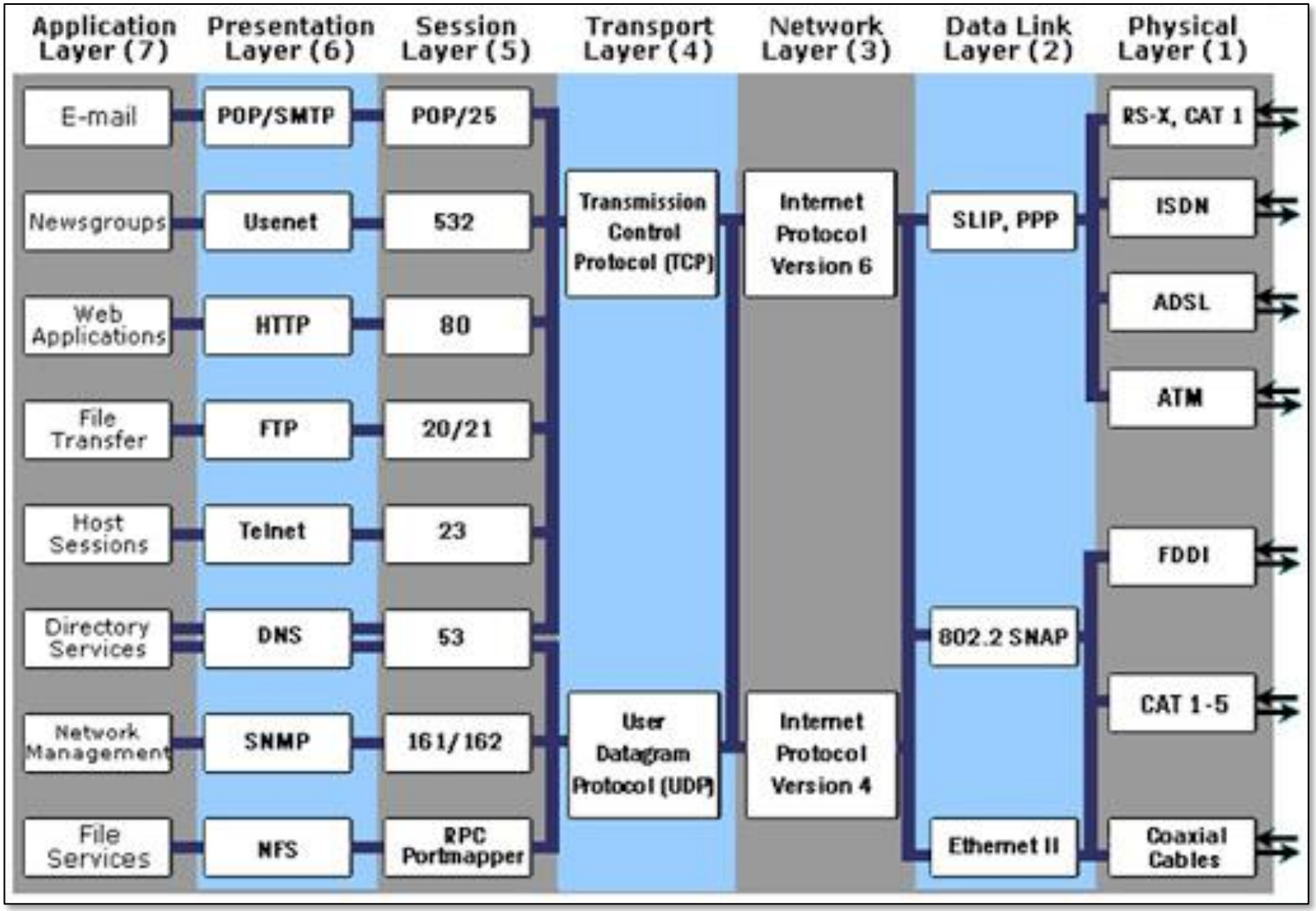
예) `{tcp, 163.239.130.90, 1500, 163.239.1.1, 3444}`

2. TCP/IP 프로토콜

- TCP/IP Protocol Family and OSI 7 Layers

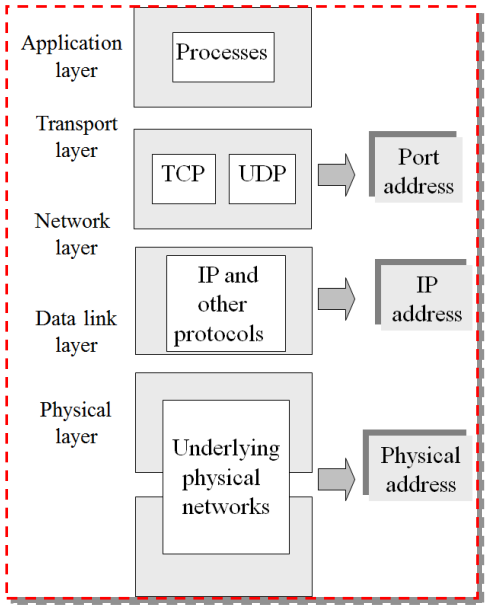


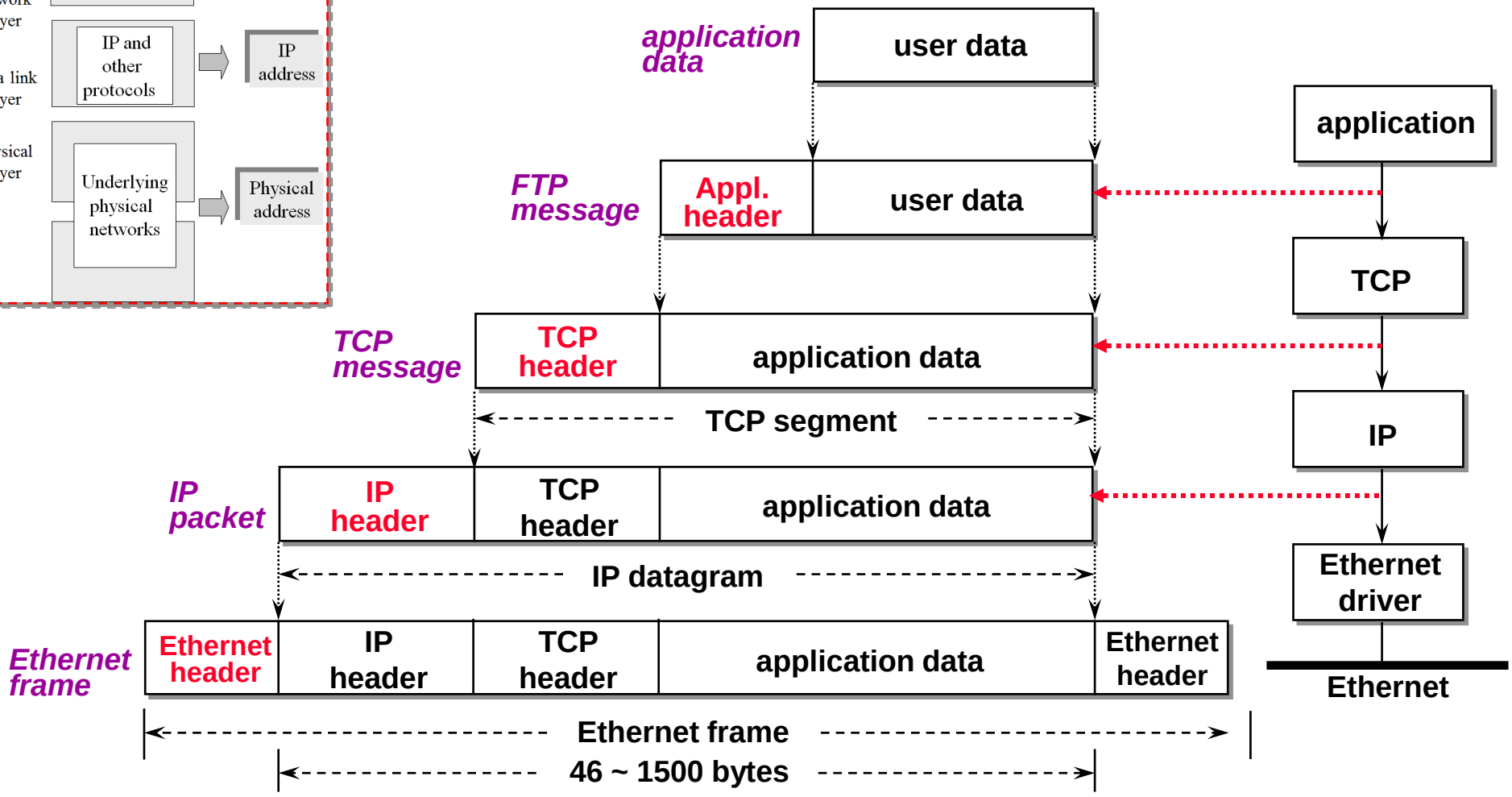


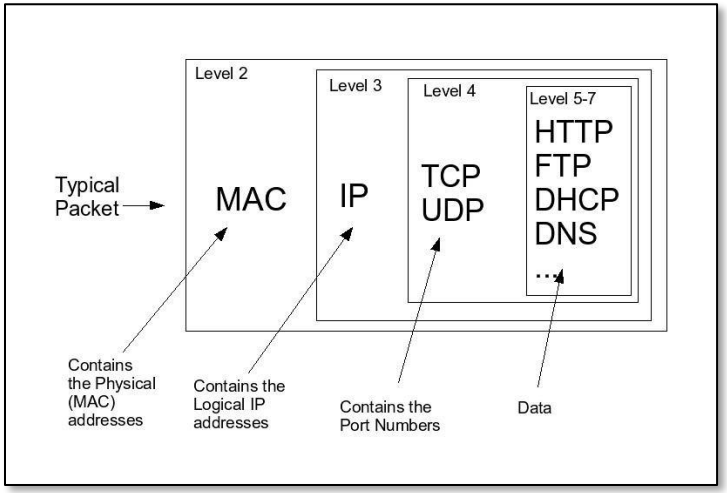

Application Layer (7)
Presentation Layer (6)
Session Layer (5)
Transport Layer (4)
Network Layer (3)
Data Link Layer (2)
Physical Layer (1)
E-mail
POP/SMTP
POP/25
Newsgroups
Usenet
532
Web Applications
HTTP
80
File Transfer
FTP
20/21
Host Sessions
Telnet
23
Directory Services
DNS
53
Network Management
SNMP
161/162
File Services
NFS
RPC Portmapper
Transmission Control Protocol (TCP)
User Datagram Protocol (UDP)
Internet Protocol Version 6
Internet Protocol Version 4
SLIP, PPP
802.2 SNAP
Ethernet II
RS-X, CAT 1
ISDN
ADSL
ATM
FDDI
CAT 1-5
Coaxial Cables

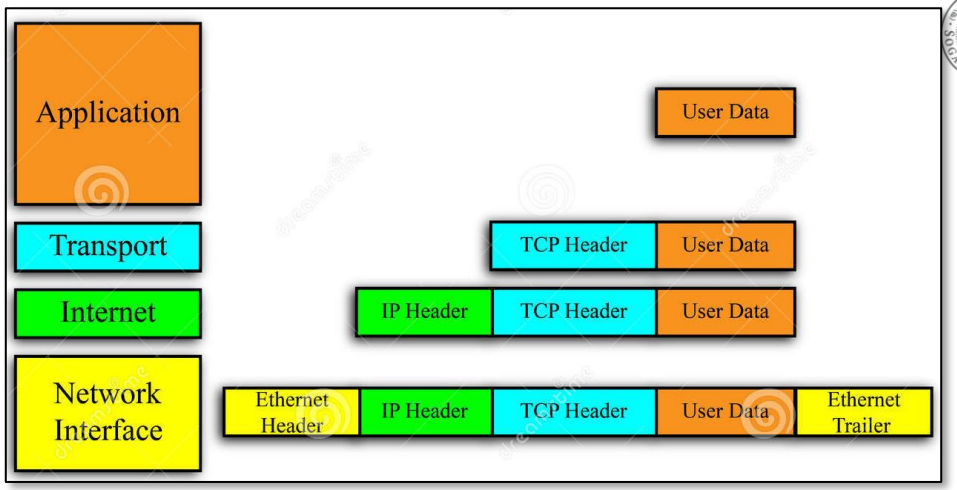
• Packet Encapsulation

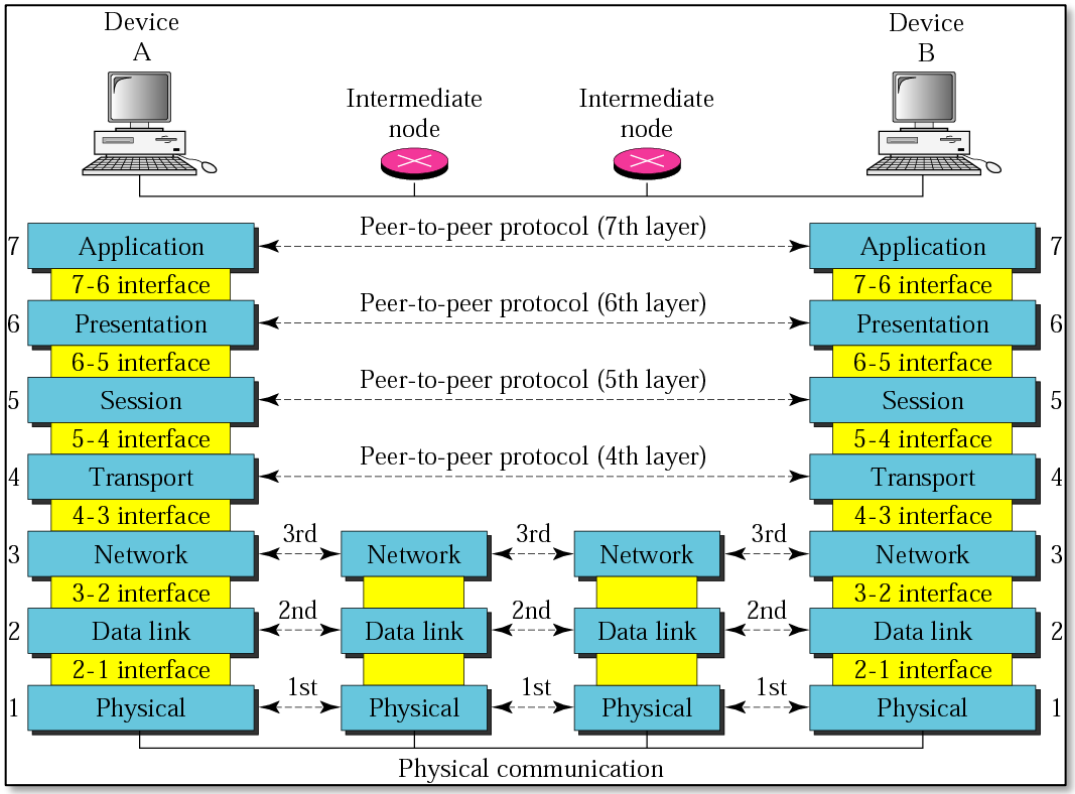
```
C:\> ipconfig /all
```

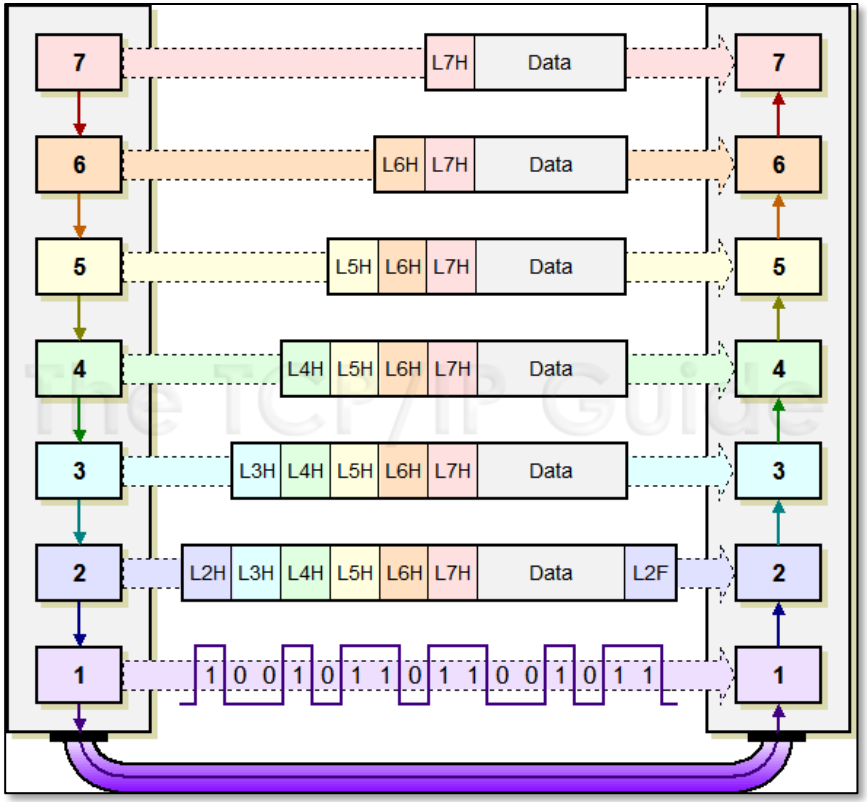





Level 2
Level 3
Level 4
Level 5-7
HTTP
FTP
DHCP
DNS
...
Typical Packet
MAC
IP
TCP
UDP
Contains the Physical (MAC) addresses
Contains the Logical IP addresses
Contains the Port Numbers
Data


Application
Transport
Internet
Network Interface
User Data
TCP Header
User Data
IP Header
TCP Header
User Data
Ethernet Header
IP Header
TCP Header
User Data
Ethernet Trailer


Device A
Device B
Intermediate node
Intermediate node
Peer-to-peer protocol (7th layer)
Peer-to-peer protocol (6th layer)
Peer-to-peer protocol (5th layer)
Peer-to-peer protocol (4th layer)
7 Application
7-6 interface
6 Presentation
6-5 interface
5 Session
5-4 interface
4 Transport
4-3 interface
3 Network
3-2 interface
2 Data link
2-1 interface
1 Physical
3rd
2nd
1st
Physical communication


7
6
5
4
3
2
1
L7H
Data
L6H
L5H
L4H
L3H
L2H
L2F
1 0 0 1 0 1 1 0 1 1 0 0 1 0 1 1

송신측(Encapsulation)			데이터 송/수신을 위한 변환과정	수신측(Decapsulation)	
내용설명(예)	계층명	프로토콜		계층명	프로토콜
사용자 네트워크 연결 (예-우편발송을 결정)	7계층[사용자 환경] (응용계층)	HTTP,SMTP,FTP Telnet		7계층(응용계층)/사용자 (예-인터넷/우편물읽음)	TTP,SMTP,FTP Telnet
정보표현 형식변환 (예-편지작성)	6계층 (표현계층)	NFS, Netbios (데이터)		6계층 (표현계층) (예-우편물 개봉)	XDR(External Data-Representation) (데이터)
상호대화 확인 (예-우편발송 접수)	5계층 (세션계층)	프로토콜 없음 (데이터)		5계층(세션계층) (예- 수취인에게 배달)	프로토콜 없음 (데이터)
상호신뢰성 확인 (예-우편등기로 결정)	4계층 (트랜스포트계층)	TCP,UDP (세그먼트)		4계층(트랜스포트계층) (예- 수취인에게 배달)	TCP,UDP (세그먼트)
최적전송로 탐색 (예- 수신자 작성)	3계층 (네트워크계층)	IP,ICMP,(R)ARP (패킷(Packet)		3계층(네트워크계층) (예- 수취인에게 배달)	IP,ICMP,(R)ARP (패킷(Packet)
전송에러 체크 (예-주소/우편번호체크)	2계층 (데이터링크계층)	HDLC,SDLC,PPP (프레임(Frame))		2계층(데이터링크계층) (예-주소/우편번호 체크)	HDLC,SDLC,PPP (프레임(Frame))
전기,물리적 발송 (예-우체부 발송)	1계층 (물리계층)	RS232C, X.24 (비트(Bit))		1계층(물리계층) (예-우편물 도착)	네트워크 장치 (비트(Bit))



Encapsulation

※ 패킷마다 송신자와 수신자 주소가 기록된후 발송.!!!

[한단계 아래로 내려갈때 마다 헤더가 하나씩 추가된다.]



⇦ 비트(Bit)신호로 전송(0101010111) ⇨

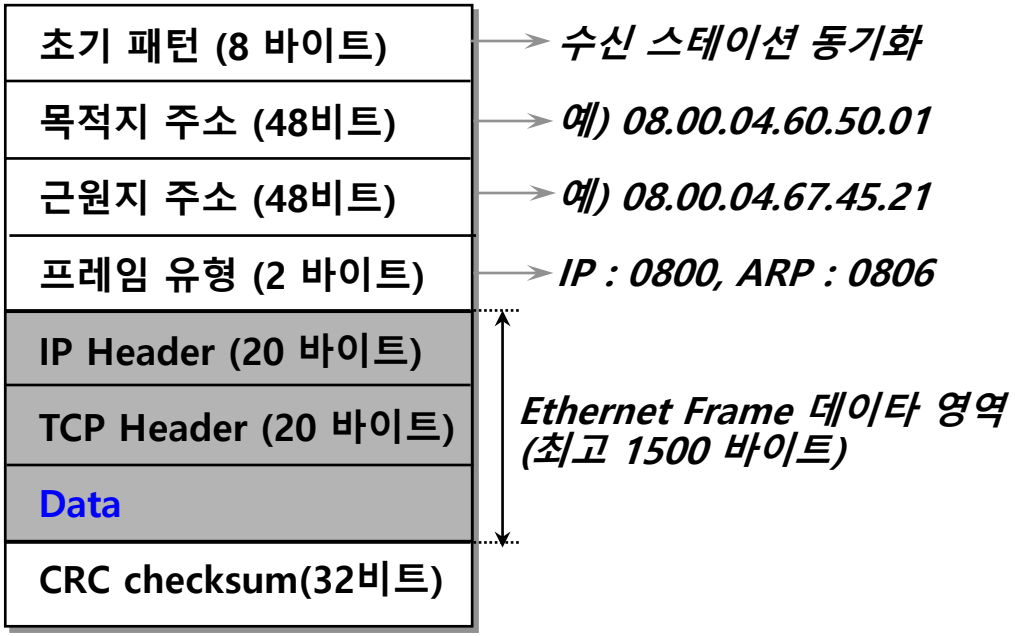
Decapsulation

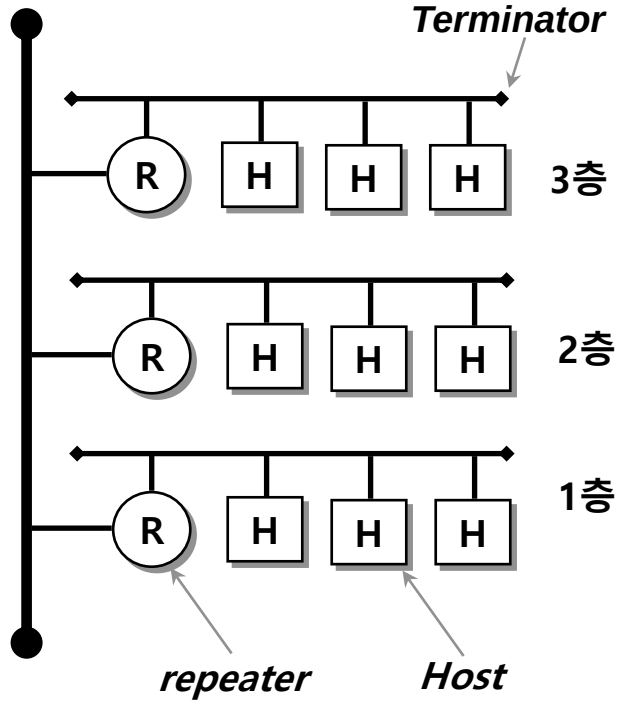
※라우터가 수신자 주소를 보고 목적지까지 전달!!

[한단계 올라올때 마다 헤더가 하나씩 벗겨진다.]

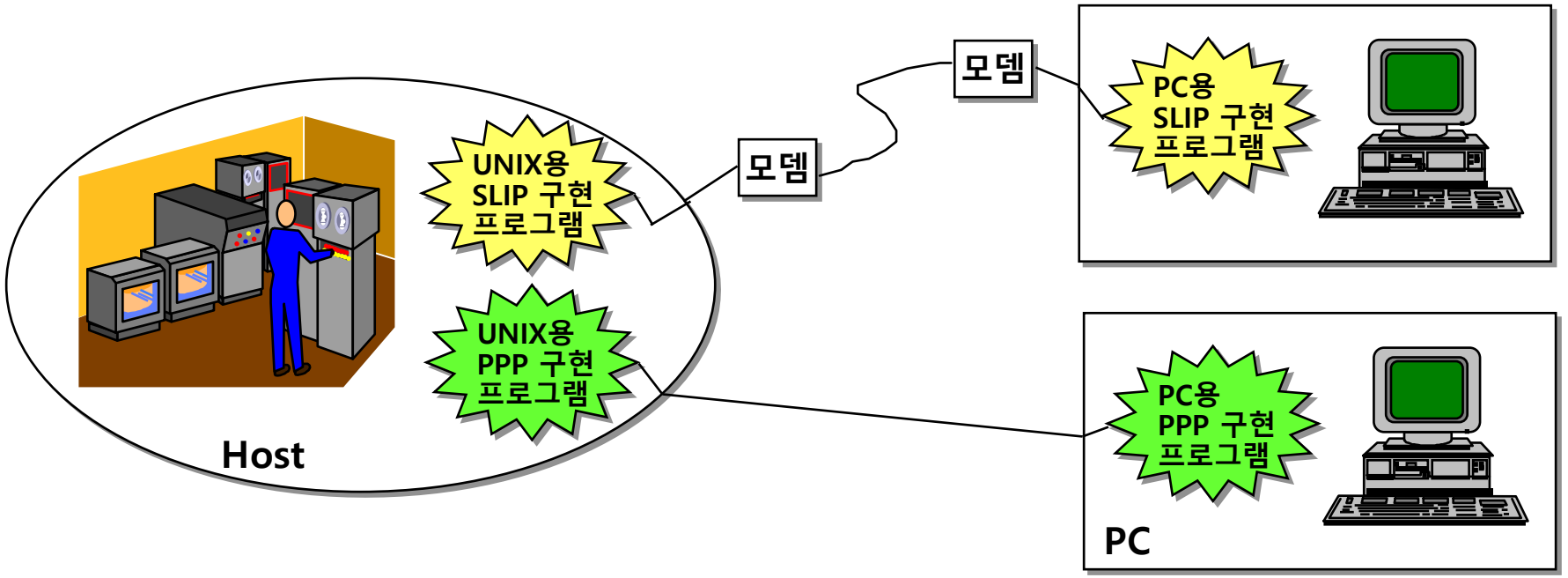
(1) 하위 네트워크 프로토콜

- **Ethernet (IEEE802.3)**
  - 1970년대초에 XEROX에서 개발한 근거리 망으로서, 현재 가장 많이 사용됨
    - ⇔ 버스로 사용되는 동축 케이블 (Coaxial Cable)의 전송 용량은 10Mbps
    - ⇔ 하나의 케이블 조각은 500미터가 최고, 200개까지의 transceiver 부착 가능
  - CSMA/CD (Carrier Sense Multiple Access with Collision Detection)
  - 각 Ethernet controller 마다 48 bit 주소가 하드웨어로 고정됨 (Ethernet Address)
    - ⇔ 예) 08.00.04.60.50.01
  - TCP/IP를 사용하는 경우 Ethernet frame 구조





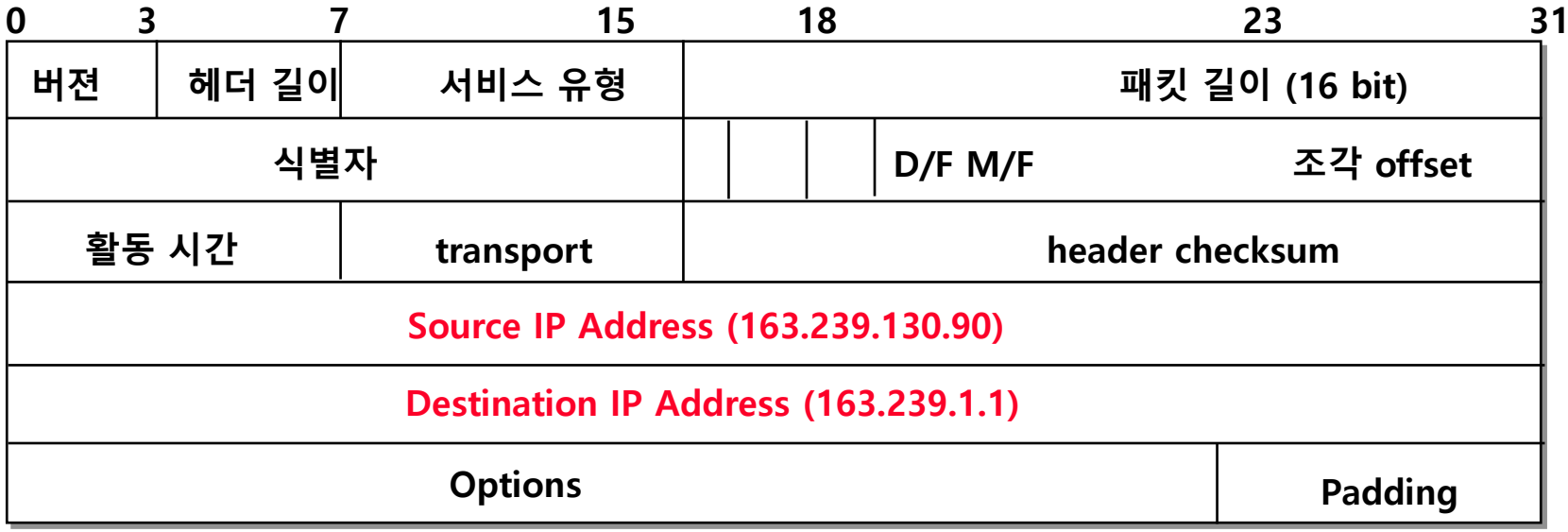
- **SLIP** (Serial Line Interface Protocol), **PPP** (Point-to-Point Protocol)
  - RS-232 serial 라인상에서 IP datagram을 전송하기 위한 프로토콜



- **X.25**
  - a point-to-point protocol
  - CCITT 표준
  - IP datagram 전송시 X.25의 최대 패킷 길이보다 더 큰 datagram이라 할지라도 IP가 패킷을 절단하지 않고 전송하도록 한다.

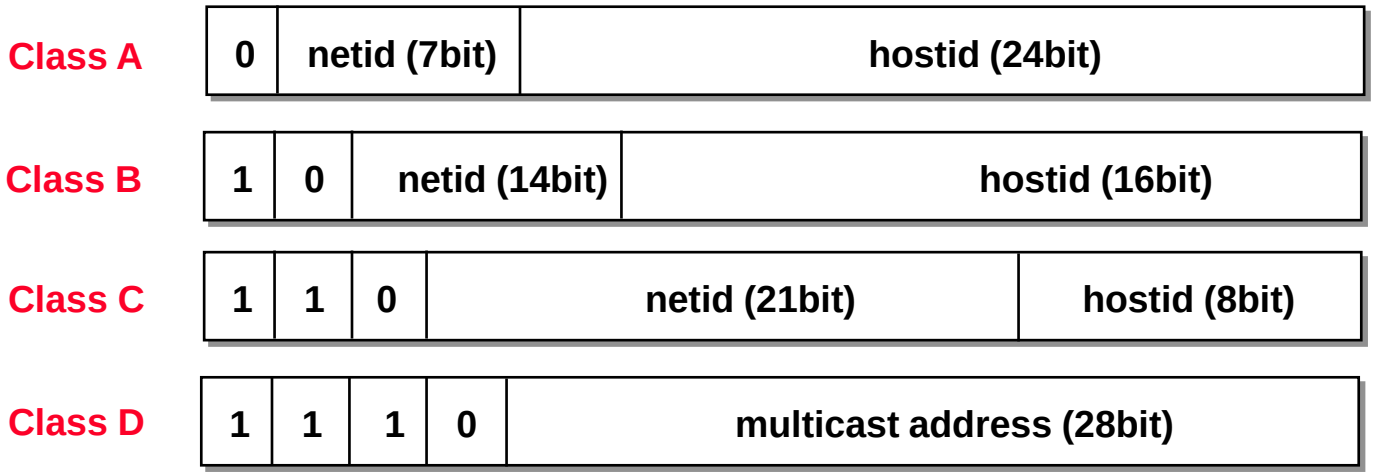
(2) IP (Internet Protocol)

- IP 특성
  - Unreliable connectionless protocol
    ⇔데이타를 제외한 header 부분에 대하여서만 checksum
  - 필요시에는 패킷을 절단 (fragmentation)
  - 32비트 인터넷 주소를 이용하여 주소 지정
  - 최대 패킷 크기 65535(2^{16}) 바이트
  - 패킷 활동 시간 제한
- IP Header

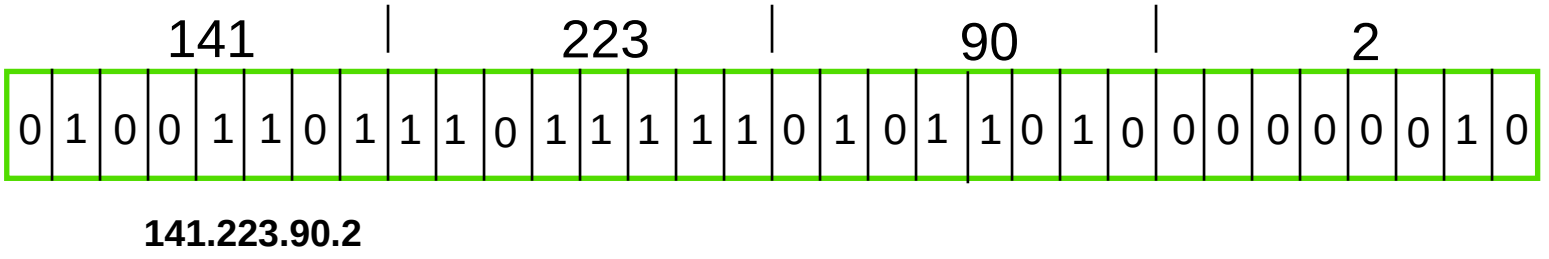


- **IP Address**
  - every host on a TCP/IP internet must have a unique 32-bit address (by NIC)
  - 32 bit addressing
    ⇔ 4,294,967,296 (2^{32})개의 호스트를 지정할 수 있음
  - 4개의 숫자와 점으로 표현
    ⇔ 예) 163.239.1.1
  - Classes

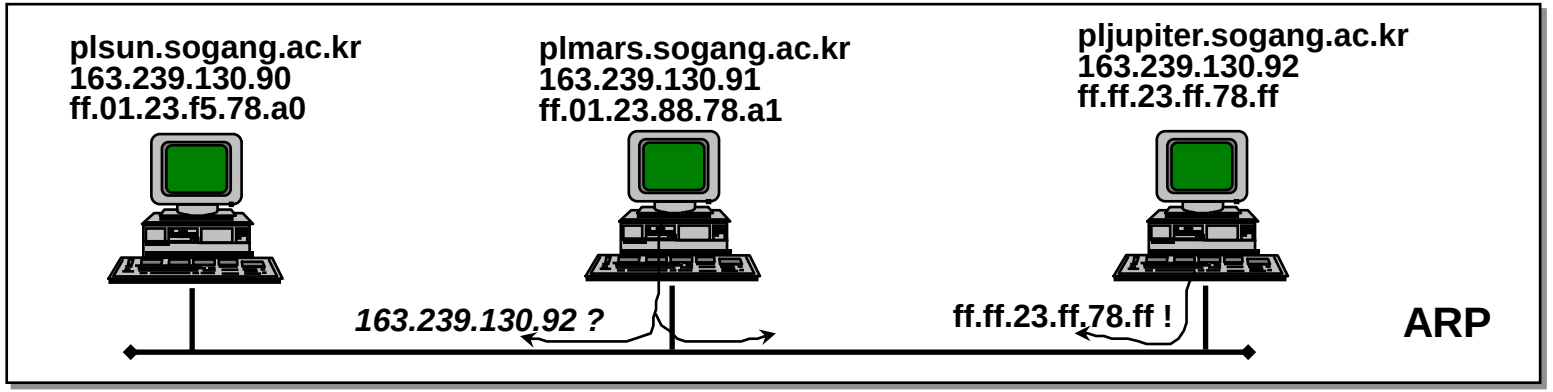
IPV6 : 128 bit addressing
$2^{128} \approx 3.4*10^{38}$



⇔ 예)

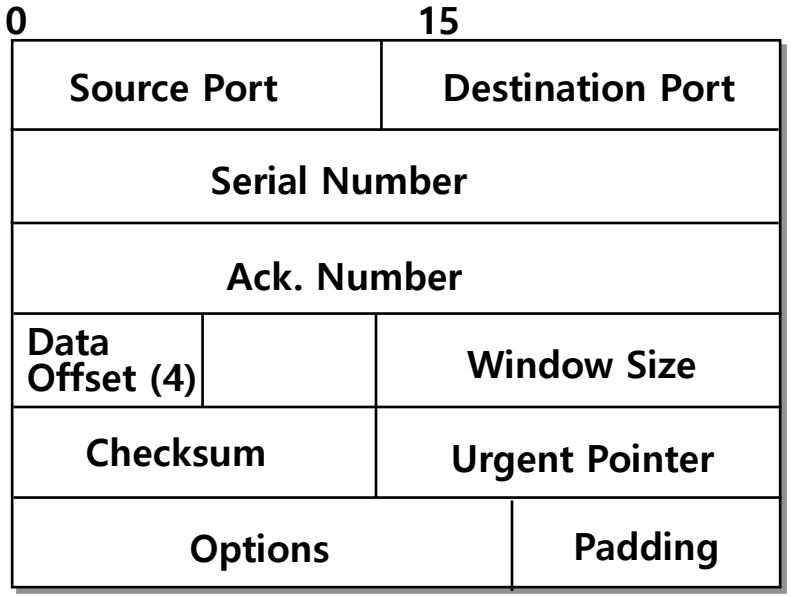


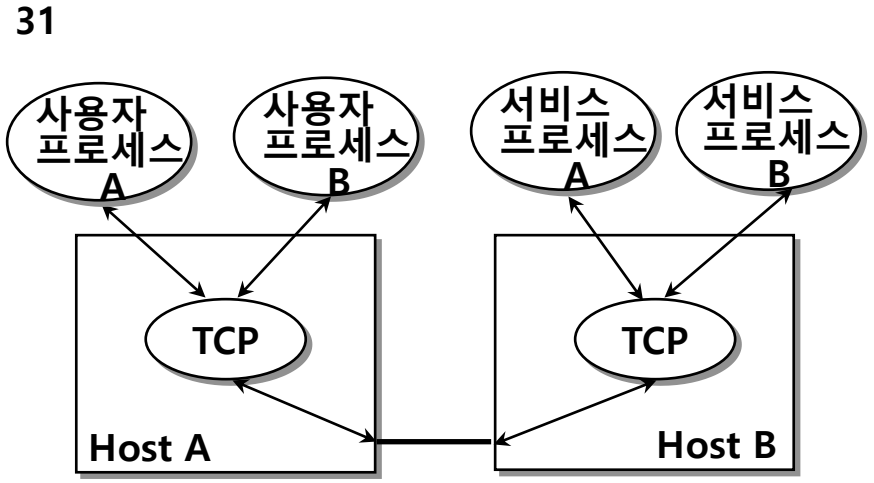
- **Address Resolution**
  - How does the IP layer determine which Ethernet address corresponds to the host with specific IP address ?
  - *ARP (Address Resolution Protocl)* : IP address -> Ethernet address
    - ⇔ a host broadcasts a special packet on the Ethernet that asks the host with a specified IP address to respond with its Ethernet address
    - ⇔ every hosts on the Ethernet receive this broadcast packet, but only the specified host should respond
  - *RARP (Reverse ARP)* : Ethernet address -> IP address
    - ⇒ diskless workstation : knows its Ethernet adress, but does not know its IP address
    - ⇔ when the workstation is initialized, it obtain its 48-bit Ethernet address from the interface hardware and broadcasts an Ethernet RARP packet containing its Ethernet address and asking for its Internet address
    - ⇔ every host on the Ethernet LAN receive this broadcast, but only the RARP servers should respond



(3) TCP (Transmission Control Protocol)

- 특성
  - 동시에 양방향 전송 (full duplex)이 가능한 가상 선로 (virtual circuit) 제공
    ⇔ Connection Oriented
  - 사용자가 느끼는 데이타 전송은 블럭 단위의 전송이 아니라 한줄기의 데이타 흐름 (stream)과 같다.
  - 안전성 있는 데이타 전송 (오류 제어, 흐름 제어)
  - 슬라이딩 윈도우 프로토콜을 이용한 데이터 전송
  - TCP의 각 사용자는 16비트의 포트(port) 번호를 할당 받는다.
- TCP 프로토콜 Header



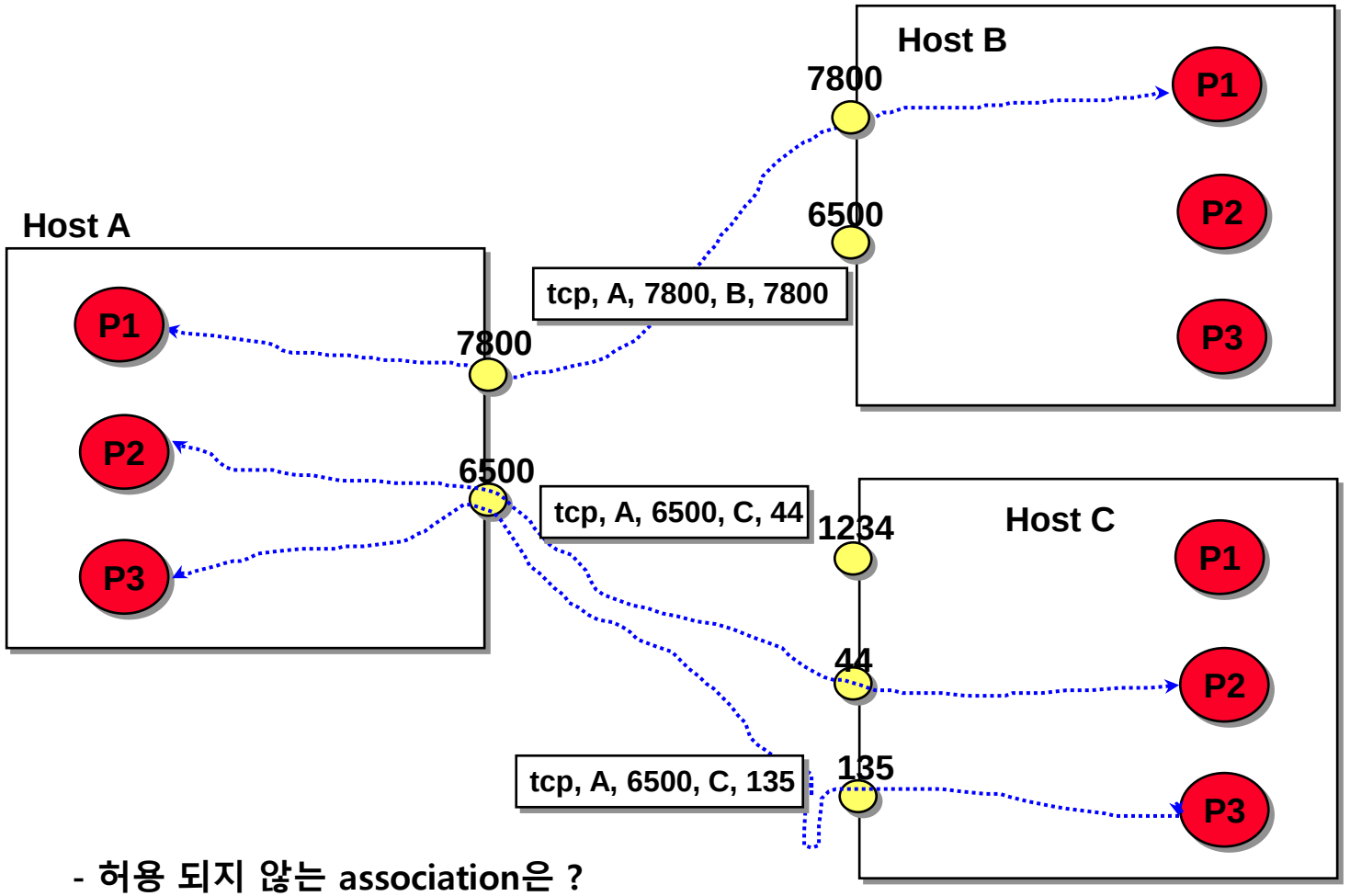


각 처리기들 사이의 통신을 어떻게 구분하는가 ?
-> Port

- **Port Number**
  - method to identify the data-connection associated with each user process
  - 16 bit를 사용
    - ⇔ 총 Port 갯수 = TCP port 갯수 (2^{16} = 65536) + UDP port 갯수 (2^{16} = 65536)
  - Association
    - ⇔ 5-tuple that defines the association
      - ⇒ the protocol (TCP or UDP)
      - ⇒ the local host Intenet address (32 bit)
      - ⇒ the local port number (16 bit)
      - ⇒ the foreign host's Internet address (32 bit)
      - ⇒ the foreign port number (16bit)
    - ⇔예) {tcp, 163.239.130.90, 1500, 163.239.1.1, 2300}
  - Well-known port
    - ⇔ how does the client identify the particular server process and its protocol on other host ?
      - ⇒ PID ? (X)
      - ⇒ Port Number ? (0)
    - ⇔ Internet well-known port and its protocol

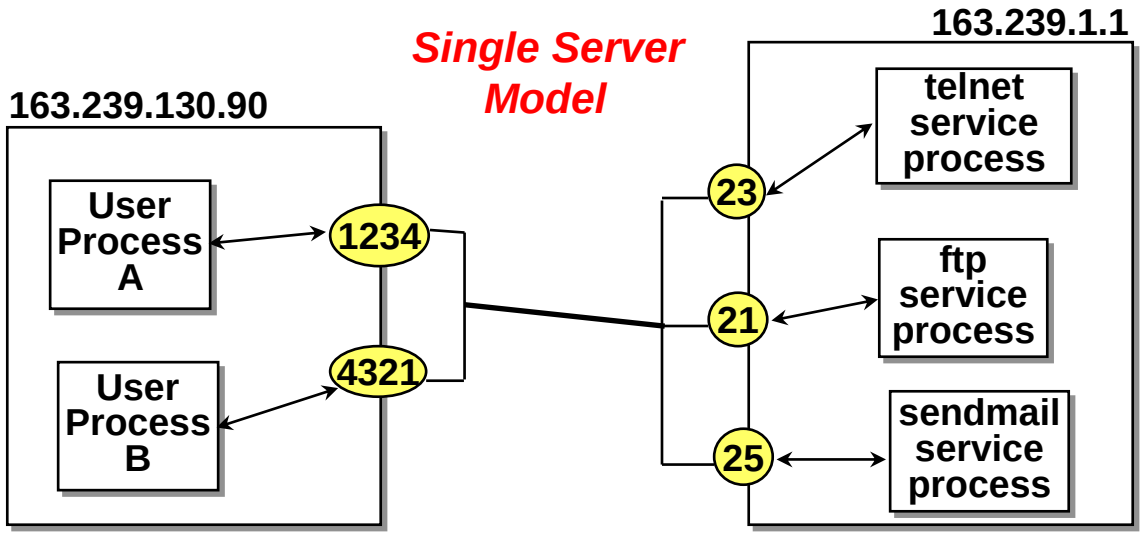
Service	Port #	Protocol
telnet	23	TCP
ftp	21	TCP
smtp	25	TCP
rlogin	513	TCP
rsh	514	TCP
portmap	111	TCP
rwhod	513	UDP
portmap	111	UDP

• Interprocess Communication using Ports

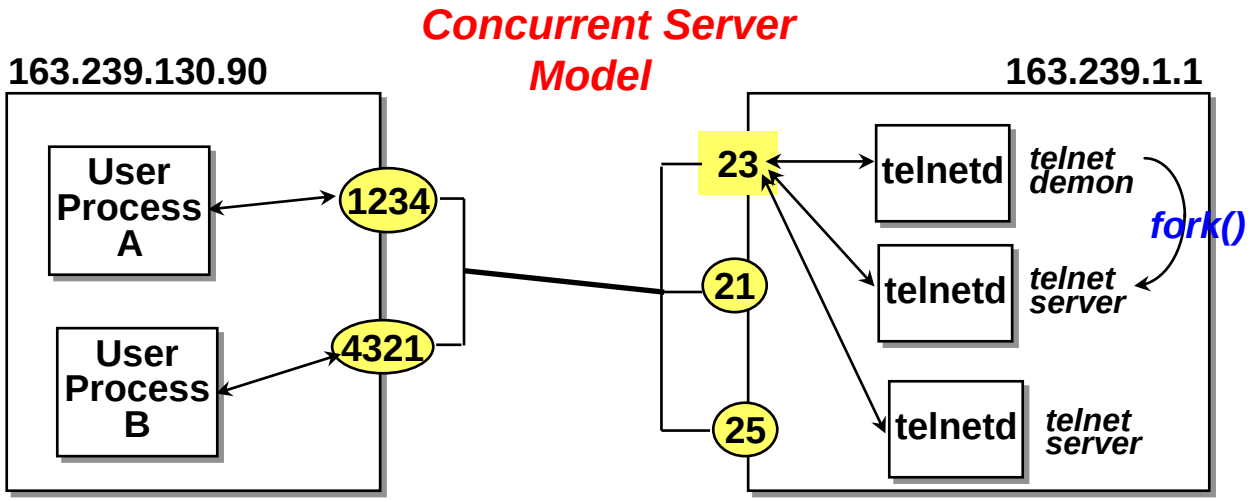


- 허용 되지 않는 association은 ?

- **TCP and Service Demon**



- associations
  {TCP, 163.239.130.90, 1234
  163.239.1.1, 23}
  {TCP, 163.239.130.90, 4321
  163.239.1.1, 21}



- associations
  {TCP, 163.239.130.90, 1234
  163.239.1.1, 23}
  {TCP, 163.239.130.90, 4321
  163.239.1.1, 23}

(4) UDP (User Datagram Protocol)

- 특성
  - 가상 선로(virtual circuit)의 개념이 없는 비연결(connectionless) 프로토콜
  - 데이타의 전송이 블럭 단위이다.
  - 안전성이 없는 데이타 전송 (블럭 재전송 및 흐름 제어 등이 없다.)
  - 슬라이딩 윈도우 등의 복잡한 기술을 사용하지 않는다.
  - UDP의 각 사용자는 16비트의 포트 번호를 할당 받는다.
- 용도
  - 안전성 있는 physical network에서의 효율성이 통신 프로그램
    ⇔ 예) NFS
- IP, TCP, UDP 특성 비교

서비스 모드	IP	TCP	UDP
connection-oriented ?	no	yes	no
message boundaries ?	yes	no	yes
data checksum ?	no	yes	opt.
positive ack. ?	no	yes	no
time out and retrmit ?	no	yes	no
duplicated detection ?	no	yes	no
sequencing ?	no	yes	no
flow control ?	no	yes	no

3. TCP/IP 관련 시스템 화일 및 관리 명령어들

- TCP/IP 관련 화일들
  - ⇔ /etc 디렉토리에 저장되어 있음
  - /etc/hostname.le0
    - ⇔ 호스트의 이름이 저장됨
  - /etc/services
    - ⇔ defines well-known ports and protocols

/etc/hostname.le0

```
pljuno
```

```
# Network services, Internet style
#
tcpmux          1/tcp           # rfc-1078
echo            7/tcp
echo            7/udp
discard         9/tcp           sink null
discard         9/udp           sink null
systat          11/tcp          users
daytime         13/tcp
daytime         13/udp
netstat         15/tcp
chargen         19/tcp          ttytst source
chargen         19/udp          ttytst source
ftp-data        20/tcp
ftp             21/tcp
telnet          23/tcp
smtp            25/tcp          mail
time            37/tcp          timserver
time            37/udp          timserver
.......
```

– /etc/hosts

⇔ IP 주소와 Internet 주소의 관계 등록

```
#****************************************
#   Sogang University, Hosts file
#   July  4, 1996
#****************************************
127.1            localhost
163.239.1.1      ccs      ccs.sogang.ac.kr
# Computer Center main hosts
163.239.1.1      ccslip   ccslip.sogang.ac.kr
163.239.1.2      ccy932   ccy932.sogang.ac.kr
#  sogang gateway at ccs
163.239.1.10     sogang-fddi-1
#  dari (kaist) - ccs (sogang) gateway at kte
128.134.1.15     dari-ccs.kaist.ac.kr

163.239.1.11     ccsun1   ccsun1.sogang.ac.kr      archie
163.239.1.12     ccsun2   ccsun2.sogang.ac.kr      news
163.239.1.13     ccsun3   ccsun3.sogang.ac.kr

163.239.130.90   plsun            plsun.sogang.ac.kr
163.239.130.91   plcupid          plcupid.sogang.ac.kr
163.239.130.92   plmars           plmars.sogang.ac.kr
163.239.130.93   plearth          plearth.sogang.ac.kr
163.239.130.94   plvenus          plvenus.sogang.ac.kr
163.239.130.95   plstar           plstar.sogang.ac.kr
163.239.130.96   plpluto          plpluto.sogang.ac.kr
163.239.130.97   pljuno           pljuno.sogang.ac.kr      loghost
```

- 관리 명령어
  - hostname
    - ⇔ 현재 시스템의 호스트 이름을 output
      혹은 변경 가능 (by root)

```
[pljuno:etc 14 ] hostname
pljuno
[pljuno:etc 15 ] hostname lala
sethostname: Not owner
[pljuno:etc 16 ] su
Passwd:
# cat /etc/hostname.le0
pljuno
# hostname plkaka
# hostname
plkaka
# exit
[plkaka:etc 17 ] _
```

  - netstat
    - ⇔ TCP/IP에 대한 다음과 같은 정보 제공
      - ⇒ 연결되어 있는 종단간 호스트의
        정보 (두 호스트 이름, 사용 포트 번호)
      - ⇒ 네트워크 접속 하드웨어에 따른 정보
      - ⇒ 데이타 버퍼
      - ⇒ 프로토콜별 통계

```
[ccs.sogang.ac.kr:jhnang 8 ] netstat -a
Active Internet connections (including servers)
Proto Recv-Q Send-Q  Local Address           Foreign Address         (state)
tcp        0      0  ccs.telnet              rheol1.1196             ESTABLISHED
tcp        0      0  ccs.3366                kids.kotel.co.kr.smtp   SYN_SENT
tcp        0      0  ccs.pop-3               chosyn1.1185            TIME_WAIT
tcp        0      0  ccs.cfinger             ccs.3365                TIME_WAIT
tcp        0      0  ccs.telnet              cryst1.1025             ESTABLISHED
tcp        0      0  ccs.3281                blue.nowcom.co.k.telne  ESTABLISHED
tcp        0    745  ccs.telnet              ccslab97.1146           ESTABLISHED
tcp        0      0  ccs.telnet              phquan1.1036            ESTABLISHED
tcp        0      0  ccs.telnet              ccs-gw.9669             ESTABLISHED
tcp        0      0  *.smtp                  *.*                     LISTEN
tcp        0      0  ccs.telnet              ccs-gw.9652             ESTABLISHED
tcp     4096      0  ccs.telnet              ccs-gw.9613             ESTABLISHED
tcp        0      0  ccs.20                  biol3.1024              ESTABLISHED
tcp        0      0  ccs.ftp                 biol3.1279              ESTABLISHED
tcp        0   1024  ccs.3356                kumoh.kumoh.ac.k.smtp   ESTABLISHED
tcp        0      0  *.domain                *.*                     LISTEN
tcp        0      0  ccs.telnet              eeca2.1345              ESTABLISHED
tcp        0      0  ccs.telnet              ccsun4.1216             ESTABLISHED
tcp        0      0  ccs.login               graphlab.1021           ESTABLISHED
tcp        0      0  ccs.telnet              ccsnet1.1214            ESTABLISHED
tcp        0      0  ccs.smtp                203.247.180.20.2848     SYN_RCVD
tcp        0      0  ccs.finger              md2.md.chula.ac..3818   ESTABLISHED
tcp        0      0  ccs.telnet              ccs-gw.3861             ESTABLISHED
tcp        0      0  ccs.1550                sparcserver.lrz-.smtp   ESTABLISHED
tcp        0      0  *.807                   *.*                     LISTEN
tcp        0      0  *.printer               *.*                     LISTEN
tcp        0      0  *.910                   *.*                     LISTEN
udp        0      0  *.domain                *.*
udp        0      0  ccs.domain              *.*
udp        0      0  localhost.domain        *.*
udp        0      0  *.1021                  *.*
```

Show all sockets

– ping & traceroute

⇔ ping : 어떤 호스트가 현재 사용 가능한지를 check한다.

⇔ traceroute : route되는 경로를 보여준다. https://traceroute-online.com/

```
[pljuno:jhnang 5 ] ping www.microsoft.com
www.microsoft.com is alive

[pljuno:jhnang 6 ] ping -s www.microsoft.com
PING www.microsoft.com: 56 data bytes
64 bytes from microsoft.com (207.68.137.7): icmp_seq=0. time=830. ms
64 bytes from microsoft.com (207.68.137.7): icmp_seq=1. time=1492. ms
64 bytes from microsoft.com (207.68.137.7): icmp_seq=2. time=1020. ms
64 bytes from microsoft.com (207.68.137.7): icmp_seq=3. time=1021. ms
64 bytes from microsoft.com (207.68.137.7): icmp_seq=4. time=1022. ms
64 bytes from microsoft.com (207.68.137.7): icmp_seq=5. time=1020. ms
^C
----7.137.68.207.in-addr.arpa PING Statistics----
8 packets transmitted, 6 packets received, 25% packet loss
round-trip (ms)  min/avg/max = 830/1067/1492

[pljuno:jhnang 6 ] traceroute www.stanford.edu
traceroute to www.stanford.edu (36.190.0.136), 30 hops max, 40 byte packets
 1  sogang-fddi-1.sogang.ac.kr (163.239.1.10)  2 ms  2 ms  2 ms
 2  t1-kt-sogang-univ.kornet.nm.kr (168.126.25.182)  4 ms  8 ms  4 ms
 3  168.126.61.190 (168.126.61.190)  22 ms  4 ms  4 ms
 4  seoul-center4-fddi2/0.kornet.nm.kr (168.126.63.44)  5 ms  4 ms  5 ms
 5  gsl-stock-1-Serial3/4.gsl.net (204.59.128.81)  449 ms  449 ms  628 ms
 6  * * gsl-sl-stock-fddi.gsl.net (204.59.128.198)  643 ms
 7  sl-stk-6-F0/0.sprintlink.net (144.228.40.6)  657 ms  688 ms  709 ms
 8  sl-mae-w-H1/0-T3.sprintlink.net (144.228.10.46)  693 ms  648 ms  675 ms
 9  * mae-w.bbnplanet.net (198.32.136.14)  668 ms  675 ms
10  * paloalto-br2.bbnplanet.net (4.0.1.13)  658 ms  823 ms
11  su-pr1.bbnplanet.net (131.119.0.199)  697 ms  690 ms *
12  * 198.31.10.1 (198.31.10.1)  679 ms  675 ms
13  * * Core-gateway.Stanford.EDU (36.36.0.1)  669 ms
14  * ceras-gateway.Stanford.EDU (36.1.0.6)  677 ms  721 ms
15  * www.Stanford.EDU (36.190.0.136)  644 ms  708 ms
[pljuno:jhnang 7 ]
```

– inetd

⇔ 데몬 서버에 대한 서버 (super server)

⇒ 모든 서버를 대신하여 포트를 열고 연결 요청을 기다린다.

⇔ inetd의 조정 화일은 /etc/inetd.conf

```
# Internet services syntax:
#  <service_name> <socket_type> <proto> <flags> <user> <server_pathname> <args>
#
# Ftp and telnet are standard Internet services.
#
ftp     stream  tcp     nowait  root    /usr/etc/in.ftpd        in.ftpd
telnet  stream  tcp     nowait  root    /usr/etc/in.telnetd     in.telnetd

name    dgram   udp     wait    root    /usr/etc/in.tnamed      in.tnamed

shell   stream  tcp     nowait  root    /usr/etc/in.rshd        in.rshd
login   stream  tcp     nowait  root    /usr/etc/in.rlogind     in.rlogind
exec    stream  tcp     nowait  root    /usr/etc/in.rexecd      in.rexecd
comsat  dgram   udp     wait    root    /usr/etc/in.comsat      in.comsat
talk    dgram   udp     wait    root    /usr/etc/in.talkd       in.talkd

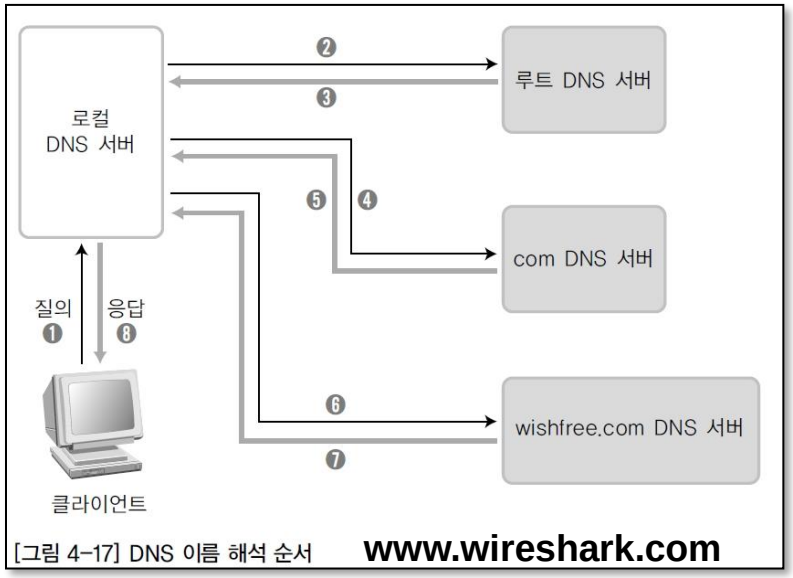
uucp    stream  tcp     nowait  root    /usr/etc/in.uucpd       in.uucpd

tftp    dgram   udp     wait    root    /usr/etc/in.tftpd  in.tftpd -s /tftpboot

finger  stream  tcp     nowait  nobody  /usr/etc/in.fingerd     in.fingerd
echo    stream  tcp     nowait  root    internal
echo    dgram   udp     wait    root    internal
discard stream  tcp     nowait  root    internal
discard dgram   udp     wait    root    internal
daytime stream  tcp     nowait  root    internal
daytime dgram   udp     wait    root    internal
chargen stream  tcp     nowait  root    internal
chargen dgram   udp     wait    root    internal
```

```
# kill -HUP <inetd PID>
#
```

• DNS (Domain Name Server)



[그림 4-17] DNS 이름 해석 순서

```
C:₩WINDOWS₩system32₩cmd.exe
C:₩>ipconfig /displaydns

Windows IP Configuration

        mrd.empas.com
        ----------------------------------------
        Record Name . . . . . : mrd.empas.com
        Record Type . . . . . : 1
        Time To Live  . . . . : 84921
        Data Length . . . . . : 4
        Section . . . . . . . : Answer
        A (Host) Record . . . : 61.97.66.16


        1.0.0.127.in-addr.arpa
        ----------------------------------------
        Record Name . . . . . : 1.0.0.127.in-addr.arpa.
        Record Type . . . . . : 12
        Time To Live  . . . . : 603881
        Data Length . . . . . : 4
        Section . . . . . . . : Answer
        PTR Record  . . . . . : localhost
```

윈도우에서 캐시된 DNS 정보 확인

```
C:₩WINDOWS₩system32₩cmd.exe - nslookup
C:₩>nslookup
Default Server:  qns1.hananet.net
Address:  210.220.163.82

> _
```

```
C:₩WINDOWS₩system32₩cmd.exe - nslookup
> www.google.co.kr
Server:  kns.kornet.net
Address:  168.126.63.1

Non-authoritative answer:
Name:    www.l.google.com
Addresses:  72.14.235.104, 72.14.235.147, 72.14.235.99
Aliases:  www.google.co.kr, www.google.com

> _
```

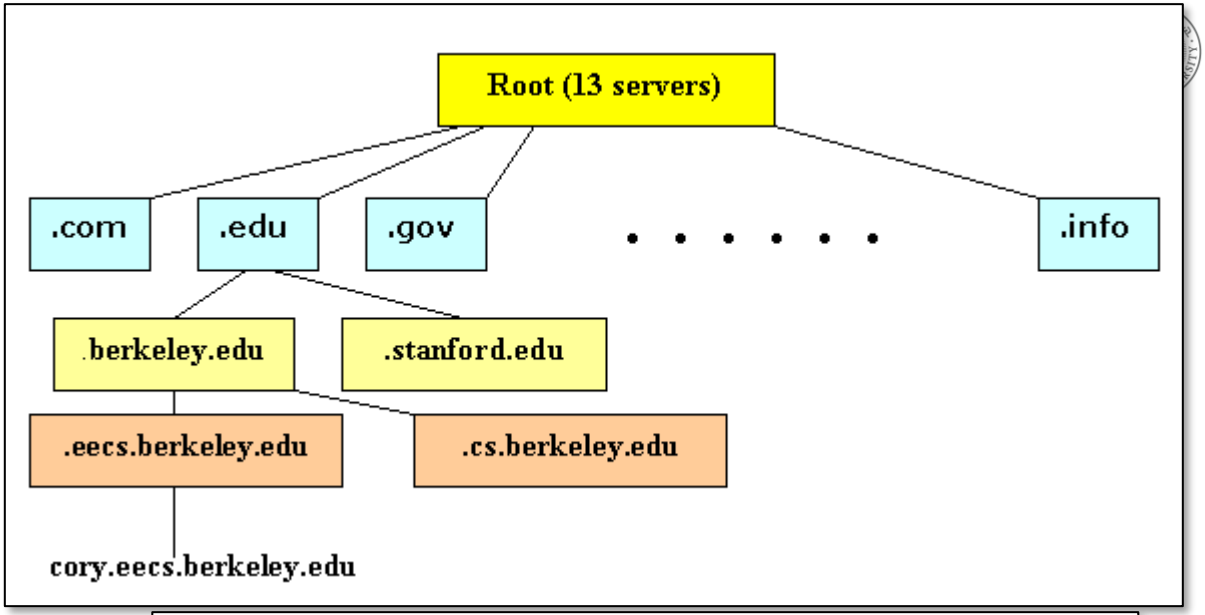
```
C:₩WINDOWS₩system32₩cmd.exe - nslookup
> set type=all
> google.co.kr
Server:  kns.kornet.net
Address:  168.126.63.1

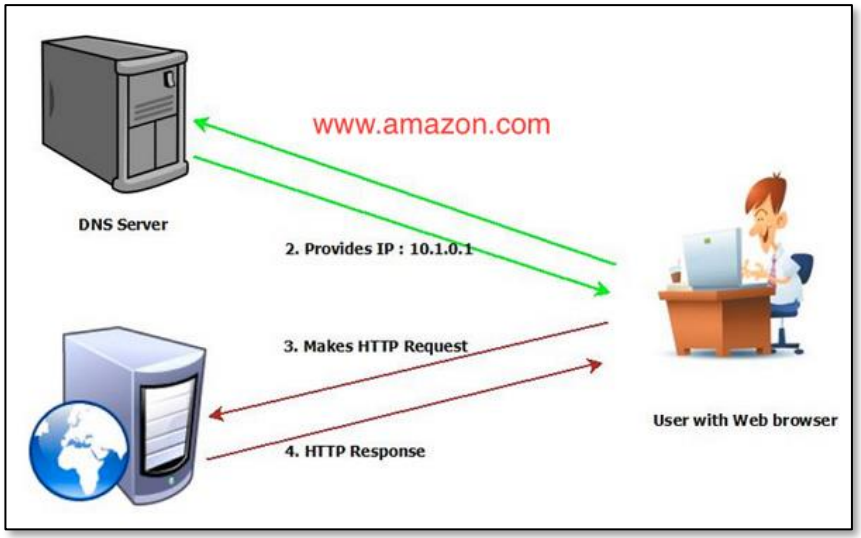
Non-authoritative answer:
google.co.kr    MX preference = 10, mail exchanger = smtp4.google.com
google.co.kr    MX preference = 10, mail exchanger = smtp1.google.com
google.co.kr    MX preference = 10, mail exchanger = smtp2.google.com
google.co.kr    MX preference = 10, mail exchanger = smtp3.google.com
google.co.kr    internet address = 209.85.171.104
google.co.kr    internet address = 72.14.207.104
google.co.kr    internet address = 72.14.235.104
google.co.kr    nameserver = ns1.google.com
google.co.kr    nameserver = ns4.google.com
google.co.kr    nameserver = ns2.google.com
google.co.kr    nameserver = ns3.google.com

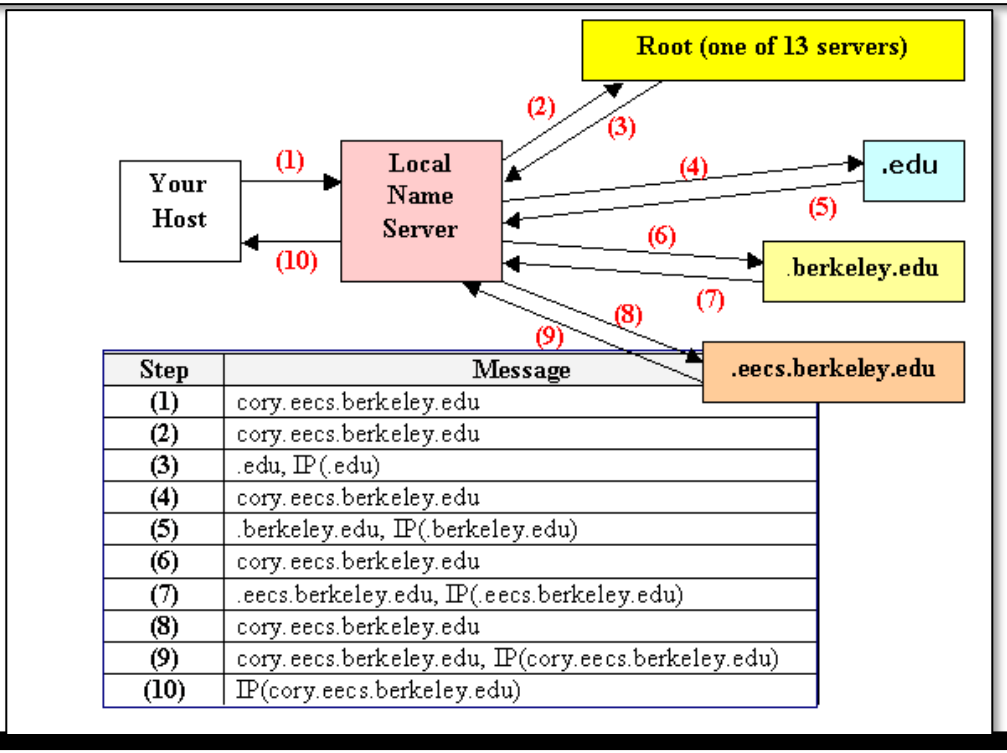
google.co.kr    nameserver = ns2.google.com
google.co.kr    nameserver = ns4.google.com
google.co.kr    nameserver = ns3.google.com
google.co.kr    nameserver = ns1.google.com
smtp1.google.com        internet address = 209.85.237.25
smtp2.google.com        internet address = 64.233.167.25
smtp3.google.com        internet address = 64.233.183.25
smtp4.google.com        internet address = 72.14.221.25
ns1.google.com  internet address = 216.239.32.10
ns2.google.com  internet address = 216.239.34.10
ns3.google.com  internet address = 216.239.36.10
ns4.google.com  internet address = 216.239.38.10
>
```

google.co.kr 도메인에 등록된 모든 서버 보기

DNS Service







Step	Message
(1)	cory.eecs.berkeley.edu
(2)	cory.eecs.berkeley.edu
(3)	.edu, IP(.edu)
(4)	cory.eecs.berkeley.edu
(5)	.berkeley.edu, IP(.berkeley.edu)
(6)	cory.eecs.berkeley.edu
(7)	.eecs.berkeley.edu, IP(.eecs.berkeley.edu)
(8)	cory.eecs.berkeley.edu
(9)	cory.eecs.berkeley.edu, IP(cory.eecs.berkeley.edu)
(10)	IP(cory.eecs.berkeley.edu)